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The Coversheet

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Table of Contents

Est. 1841 | YORK ST JOHN UNIVERSITY

Table of Figures

List of Tables

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Exploring the Socioeconomic Determinants of Poverty Across the United States from 2018 to 2022: A Data Analysis and Visualisation Project

LDS7004M Data Visualisation

Assessment Report

1.0 Introduction

1.1 Background

Poverty remains a significant socioeconomic challenge in the United States, affecting millions of households and influencing access to essential services such as housing, childcare, education, and healthcare (U.S. Census Bureau, 2023). While poverty rates vary considerably across states, the factors contributing to these differences are often interconnected and complex. Understanding the underlying drivers of poverty is therefore important for policymakers, researchers, and social welfare organisations seeking to develop targeted interventions that improve economic wellbeing.

The primary objective of this analysis is to identify the key factors associated with variations in poverty levels across U.S. states, for example, one factor that has attracted increasing attention in recent years is the affordability of childcare. For many families, childcare represents a substantial household expense that directly affects employment decisions, labour force participation, and disposable income. Rising childcare costs may place additional financial pressure on low-income households, potentially increasing the risk of economic hardship and poverty.

To achieve the set objective, the project employs exploratory data analysis (EDA) techniques alongside multivariate visualisations to investigate relationships between poverty rates and other socioeconomic indicators. Attention is given to identifying variables that exhibit strong associations with poverty and assessing how these relationships differ across states and analysing some trends over time.

Data visualisation serves as the central analytical approach throughout the project. Effective visual representations enable complex datasets to be interpreted more intuitively, revealing trends, correlations, and geographic disparities that may not be immediately apparent through numerical analysis alone. By combining statistical exploration with visual storytelling, the project aims to communicate meaningful insights into the socioeconomic conditions that contribute to poverty across the United States.

The analysis was conducted using the Python programming language, supported by several data science and visualisation libraries, including Pandas (McKinney, 2010) for data manipulation, Matplotlib (Hunter, 2007) and Seaborn (Waskom, 2021) for static data visualisations, and Plotly (Plotly Technologies Inc., 2024) for interactive graphical visualisation.

1.2 Dataset Selection

This project investigates the underlying insights associated with poverty rates across the United States using dataset derived from the National Database of Childcare Prices (NDCP). The NDCP is the most comprehensive federal source of county-level childcare pricing information and combines childcare market data with demographic and economic indicators, providing a valuable resource for socioeconomic analysis (U.S. Department of Labor, 2024). The dataset consists of 48,308 observations and 370 attributes.

For this study, data covering the period from 2018 to 2022 were utilised to examine patterns, trends, and relationships between poverty rates and selected economic and demographic variables.

2.0 Data Preparation and Processing

2.1 Data Ingestion

The open dataset was downloaded directly from the National Database of Childcare Prices (NDCP) on the U.S. Department of Labor website. Following the download, the raw data file was imported into the working environment using Pandas module for further processing and analysis.

2.2 Data Cleaning

The following subsections outline the data preprocessing and cleaning steps undertaken to ensure that only relevant variables were retained for analysis, all missing values were appropriately addressed, and any invalid or erroneous outliers were identified and handled accordingly.

2.2.1 Filtering for the Relevant Columns and Rows

A substantial proportion of these attributes are metadata fields used to indicate whether specific values were systematically imputed. For example, the variable **iemr_16_state** is a binary indicator identifying observations where the female employment rate was imputed. As these metadata variables do not contribute directly to the analytical objectives of this study and for the aim of aligning the dataset with the scope of the study, the dataset was subsequently filtered to remove them alongside other attributes deemed to have limited relevance to the investigation of poverty-related factors.

To ensure that the analysis focused exclusively on the most recent five-year period under investigation, only observations recorded between 2018 and 2022 were used.

Table 1: *List of all the relevant variables used in this analysis*

Variable Name	Description	Data Type
state_name	Name of the state	Categorical
state_abbreviation	Two-letter state abbreviation	Categorical

county_name	Name of the county	Categorical
county_fips_code	Five-digit county identifier (FIPS code)	Categorical
studyyear	Year in which the data were collected	Categorical
unr_16	Unemployment rate of individuals aged 16 years and above	Numeric
funr_16	Unemployment rate of females aged 16 years and above	Numeric
munr_16	Unemployment rate of males aged 16 years and above	Numeric
mhi_2022	Median household income (2022 USD)	Numeric
totalpop	Total population count	Numeric
h_under6_fwork	Households with children under six years old where only the father is employed	Numeric
h_under6_mwork	Households with children under six years old where only the mother is employed	Numeric
h_under6_singlem	Households with children under six years old headed by a single mother	Numeric
h_6to17_fwork	Households with children aged 6–17 years where only the father is employed	Numeric
h_6to17_mwork	Households with children aged 6–17 years where only the mother is employed	Numeric
h_6to17_singlem	Households with children aged 6–17 years headed by a single mother	Numeric
h_under6_bothwork	Households with children under six years old where both parents are employed	Numeric
h_6to17_bothwork	Households with children aged 6–17 years where both parents are employed	Numeric
mcbto5	Median weekly full-time centre-based childcare cost (0–5 months)	Numeric
mc6to11	Median weekly full-time centre-based childcare cost (6–11 months)	Numeric
mc12to17	Median weekly full-time centre-based childcare cost (12–17 months)	Numeric
mc18to23	Median weekly full-time centre-based childcare cost (18–23 months)	Numeric
mc24to29	Median weekly full-time centre-based childcare cost (24–29 months)	Numeric
mc30to35	Median weekly full-time centre-based childcare cost (30–35 months)	Numeric
mc36to41	Median weekly full-time centre-based childcare cost (36–41 months)	Numeric
mc42to47	Median weekly full-time centre-based childcare cost (42–47 months)	Numeric
mc48to53	Median weekly full-time centre-based childcare cost (48–53 months)	Numeric
mc54tosa	Median weekly full-time centre-based childcare cost (54 months to school age)	Numeric
mcsa	Median weekly full-time centre-based childcare cost (school age)	Numeric

mfccbto5	Median weekly full-time family childcare cost (0–5 months)	Numeric
mfcc6to11	Median weekly full-time family childcare cost (6–11 months)	Numeric
mfcc12to17	Median weekly full-time family childcare cost (12–17 months)	Numeric
mfcc18to23	Median weekly full-time family childcare cost (18–23 months)	Numeric
mfcc24to29	Median weekly full-time family childcare cost (24–29 months)	Numeric
mfcc30to35	Median weekly full-time family childcare cost (30–35 months)	Numeric
mfcc36to41	Median weekly full-time family childcare cost (36–41 months)	Numeric
mfcc42to47	Median weekly full-time family childcare cost (42–47 months)	Numeric
mfcc48to53	Median weekly full-time family childcare cost (48–53 months)	Numeric
mfcc54tosa	Median weekly full-time family childcare cost (54 months to school age)	Numeric
mfccsa	Median weekly full-time family childcare cost (school age)	Numeric
pr_p	Poverty rate of individuals	Numeric

By focusing only on variables and rows that are relevant to this analysis project, the total number of relevant rows are 16,103 and the number of variables is 40.

2.2.2 Handling Blank or Null Rows

An initial inspection of the dataset revealed a large number of missing values distributed across 22 attributes. These missing entries were imputed using the mean value calculated at the state level rather than applying a global column mean. This approach was adopted to preserve regional variation and avoid distortion that could arise from using a national average, particularly for states with significantly different population sizes or socioeconomic characteristics.

```
print("List of columns and their number of blanks rows:")
filtered_df.isna().sum().sort_values(ascending=False)
✓ 0.0s

List of columns and their number of blanks rows:

mfccsa          3140
mfcc24to29      3026
mfcc30to35      3026
mfcc6to11       3019
mfccbto5        3019
mfcc18to23      3018
mfcc12to17      3018
mfcc36to41      3014
```

Figure 1: *Code screenshot showing the top attributes with the highest number of blank rows*

For the remaining attributes where missing values persisted after the initial imputation step, the absence of data indicated that no county-level records were available for those numerical variables across all years within the respective states. These remaining gaps were filled using the national median for each variable to maintain consistency while reducing the influence of extreme values, which could otherwise distort the dataset if the mean were used.

```
filtered_df = filtered_df.apply(fill_blanks_with_median) #fill all remaining blank rows in each attribute with the national median of e
✓ 0.0s
```

Figure 2: *Code screenshot showing the process of using a reusable function to fill the remaining blank cells wit national averages*

2.3 Feature Engineering

2.3.1 Merging related attributes in the dataset

New variables were created by combining related attributes that were analysed collectively. For example, the variables representing the number of households where both parents were working were originally recorded across different child age-group categories. These age-specific values were therefore merged into a single variable to produce an overall measure. Some related attributes were aggregated with the **mean** and some were merged with the **sum** aggregate function depending on what was statistically appropriate.

After merging, the related attributes were no longer needed and were dropped from the dataset.

```
filtered_df["2_breadwinners_hh"] = filtered_df[["h_under6_bothwork", "h_6to17_bothwork"]].sum(axis=1).astype("int64")
filtered_df = filtered_df.drop(["h_under6_bothwork", "h_6to17_bothwork"], axis=1) #dropping the extra columns that are no more needed
filtered_df.head(1)
✓ 0.0s                                                                                      Python
```

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	...	mfcc24to29	mfcc30to35	mfcc36to41
0	Alabama	AL	Autauga County	1001	2018	4.2	3.4	4.9	69367	55200	...	111.58	110.93	110.93

1 rows × 35 columns

Figure 3*: Showing how the total number of two-breadwinner households was derived from adding two related attributes*

2.3.2 Deriving the Poverty Population Attribute

A new variable, **poverty_population**, was created to represent the total number of individuals living below the poverty line within each county. This measure provides a more concrete indication of poverty levels by accounting for population size. It also supports the computation of more reliable poverty rates at both state and national levels.

```
#multiply the total population by poverty rate in each county to find the poverty population in each county
filtered_df["poverty_population"] = (filtered_df["totalpop"] * (filtered_df["pr_p"] / 100)).astype("int64")
filtered_df[["state_name","totalpop", "pr_p","poverty_population"]].head()
```

✓ 0.0s

	state_name	totalpop	pr_p	poverty_population
0	Alabama	55200	15.4	8500
1	Alabama	55380	15.2	8417
2	Alabama	55639	15.2	8457
3	Alabama	58239	13.6	7920
4	Alabama	58761	11.4	6698

Figure 4: *Calculating the* ***poverty_population*** *attribute by multiplying the total population with the poverty rate in each County*

2.3.3 Dropping Redundant Variable

The general unemployment rate variable (unr_16) was removed from the dataset because it can be indirectly derived from the gender-specific unemployment rate variables.

```
filtered_df = filtered_df.drop("unr_16", axis=1)
filtered_df.head(1)
```

✓ 0.0s

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	funr_16	munr_16	mhi_2022	totalpop	pr_p	1_breadwinner_hh	2_b
0	Alabama	AL	Autauga County	1001	2018	3.4	4.9	69367	55200	15.4	6127	

Figure 5: *Dropping derivable attribute to reduce redundancy*

2.3.4 Finalising the Dataset

To conclude the feature engineering process, the following attributes were renamed for better clarity during analysis.

Table 2: *Renaming Applied during Data Processing*

Original Variable Name	New Variable Name
funr_16	female_unemployed_rate
munr_16	male_unemployed_rate
mhi_2022	2022_med_household_inc
pr_p	poverty_rate

Upon concluding the data processing steps, the final cleaned dataset contained 16,103 observations and 15 attributes prepared for analysis and was exported into the local memory for future usage.

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	female_unemployed_rate	male_unemployed_rate	2022_med_household_inc	population	pove
0	Alabama	AL	Autauga County	1001	2018	3.4	4.9	69367	55200	
1	Alabama	AL	Autauga County	1001	2019	3.2	4.1	68128	55380	
2	Alabama	AL	Autauga County	1001	2020	2.4	3.3	66679	55639	

Figure 6: *Preview of the final dataset.*

3.0 Performing Exploratory Data Analysis

3.1 Summary Statistics and Initial Observations

The Pandas **describe()** function was used to summarise the dataset to show the central tendency summary of each numerical variable as shown in Figure 7.

```
#describe the only the numerical data in the dataset
filtered_df.drop(["county_fips_code", "studyyear"], axis = 1).describe().T
```

	count	mean	std	min	25%	50%	75%	max
female_unemployed_rate	16103.0	5.278271	3.449434	0.000000	3.300000	4.700000	6.400000	3.900000e+01
male_unemployed_rate	16103.0	5.777284	3.692781	0.000000	3.700000	5.200000	7.000000	4.650000e+01
2022_med_household_inc	16103.0	61618.497050	17608.384925	14013.000000	50924.000000	59926.000000	69707.500000	1.709350e+05
population	16103.0	102561.790536	328441.534201	48.000000	11137.500000	25931.000000	66979.500000	1.009805e+07
poverty_rate	16103.0	15.588089	7.985098	0.000000	10.500000	14.000000	18.500000	6.710000e+01
1_breadwinner_hh	16103.0	10670.253307	37195.054185	0.000000	1022.500000	2546.000000	6606.000000	1.153355e+06
2_breadwinners_hh	16103.0	9194.269515	27997.475443	0.000000	876.000000	2126.000000	5788.500000	7.647500e+05
avg_centrebased_care_price	16103.0	151.683741	47.871146	23.558182	119.028182	144.744586	169.957273	5.238682e+02
avg_family_childcare_price	16103.0	126.636543	36.222570	44.312727	101.275455	124.335106	142.661364	4.035791e+02
poverty_population	16103.0	13765.451655	46120.342047	0.000000	1630.500000	4073.000000	10377.500000	1.615688e+06

Figure 7: *Summary statistics of all numerical variables in the dataset*

Insights from the summary statistics:

The descriptive statistics provided an initial overview of the distribution and variability of the selected variables.

- The unemployment rates for both males and females exhibited evidence of high-end extreme values. Although the mean and median unemployment rates were approximately 5%, and 75% of observations recorded unemployment rates below 7%, the maximum values exceeded 35%. While these observations may be considered outliers, they likely reflect genuine economic conditions within certain counties and were therefore retained for further analysis.

- The standard deviation of the 2022 median household income indicated substantial variation in income levels across U.S. counties, suggesting notable economic disparities between regions.

- The distribution of poverty rates also contained several extreme observations. Although 75% of counties recorded poverty rates below 18.5%, some counties exhibited poverty rates exceeding 65%. These values were preserved as they represent meaningful variations within the dataset and are relevant to the objectives of the study.

- The variability observed in both centre-based and family childcare costs was relatively high when compared with their respective mean values. This suggests substantial differences in childcare affordability across the United States.

- No invalid values were identified during the review of the summary statistics. In particular, no negative values or other implausible observations were detected within the numerical variables, indicating that the data was suitable for further analysis.

3.2 Examination of Variable Distributions and Visualising Outliers

Outliers were treated differently in this study because the objective was to examine actual socioeconomic conditions across all U.S. states, including observations with unusually high values that represent legitimate records. To assess their impact and understand the distribution of each numerical variable, histograms were used to visualise their spread and shape.

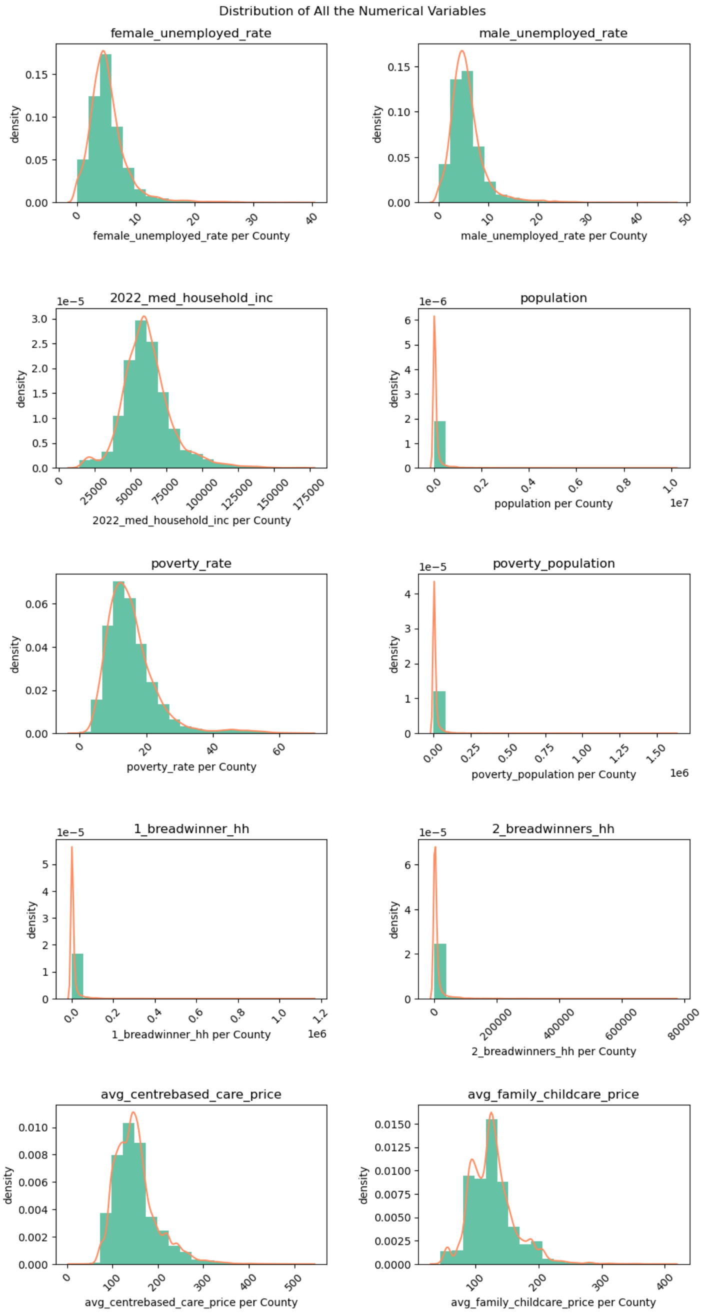

***Figure 8**: Subplots of histograms showing the spread of each numerical variable in the dataset*

The histograms and kernel density estimates indicated that most variables followed approximately normal distributions, although they all exhibited positive skewness where majority of the population are on the lower end. For example, variables such as **unemployment rates** for each gender**, poverty rate**, **population,** and **child-care price variables** were characterised by a concentration of observations at lower values, accompanied by a relatively small number of counties with exceptionally high values. These extreme observations contributed to the long right tails observed in the distributions and reflect substantial variation across U.S. counties.

3.3 Analysing the Correlation between the Numerical Variables

A correlation heatmap was selected to visualise relationships between the numerical variables in the dataset because it provides a compact and intuitive overview of pairwise associations within large multivariate data. A heatmap enables simultaneous comparison of all correlations in a single visual structure, improving interpretability and analytical efficiency. This makes it particularly useful for identifying patterns such as multicollinearity, strong positive or negative relationships, and weak or insignificant associations across socioeconomic indicators (Wilkinson and Friendly, 2009).

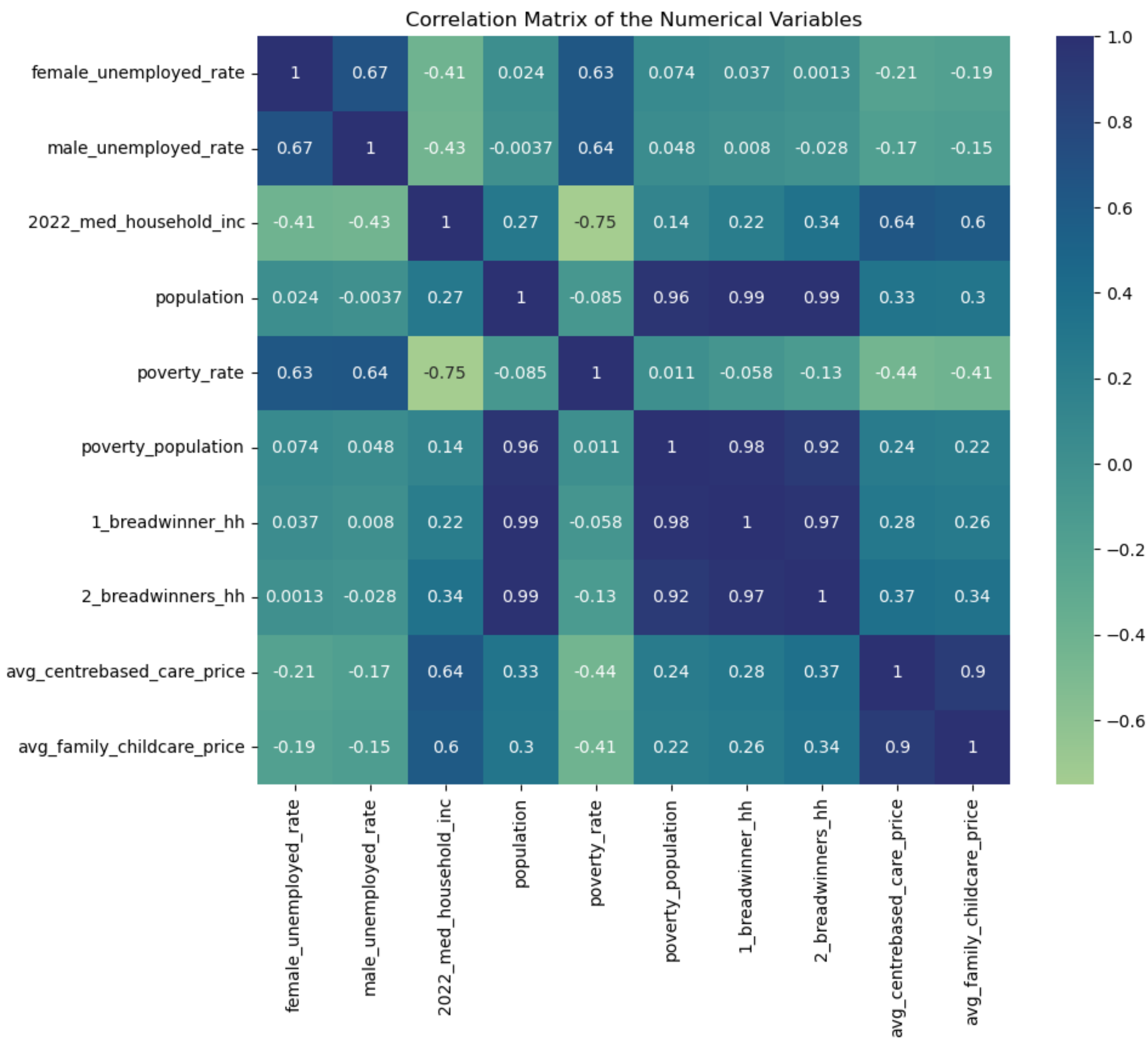

__Figure 9:__ Heatmap visualising correlation scores of all the numerical variables.

3.4 Analysing the Unemployment Rate Variable as a Key Determinant of Poverty Rate

The examination of the unemployment rate variable as a key determinant of poverty rate revealed that Guánica County recorded the highest male unemployment rate at 46.5% in 2020, while Utuado County exhibited the highest female unemployment rate at 39% in 2020. Both counties are in the State of Puerto Rico.

At the lower end, approximately thirty-five counties reported a 0% unemployment rate. A large proportion of these counties are characterised by very small population sizes.

	state_name	county_name	female_unemployed_rate	male_unemployed_rate
0	Colorado	Kiowa County	0.0	0.0
1	Hawaii	Kalawao County	0.0	0.0
2	Kansas	Greeley County	0.0	0.0
3	Kansas	Hamilton County	0.0	0.0
4	Montana	Fallon County	0.0	0.0
5	Montana	Garfield County	0.0	0.0

Figure 10: *Preview of the Counties with the lowest unemployment rates reported in the dataset in the United States*

Furthermore, by examining the full five-year trend in annual unemployment rates, the bar chart in Figure 11 illustrates a steady decline in unemployment in the United States of America between 2018 and 2022, reaching its lowest level in 2022 at 5.13%. A bar chart was chosen for this visualisation because it provided a clear and effective way of comparing discrete values across different years, allowing trends over time to be easily identified without visual clutter.

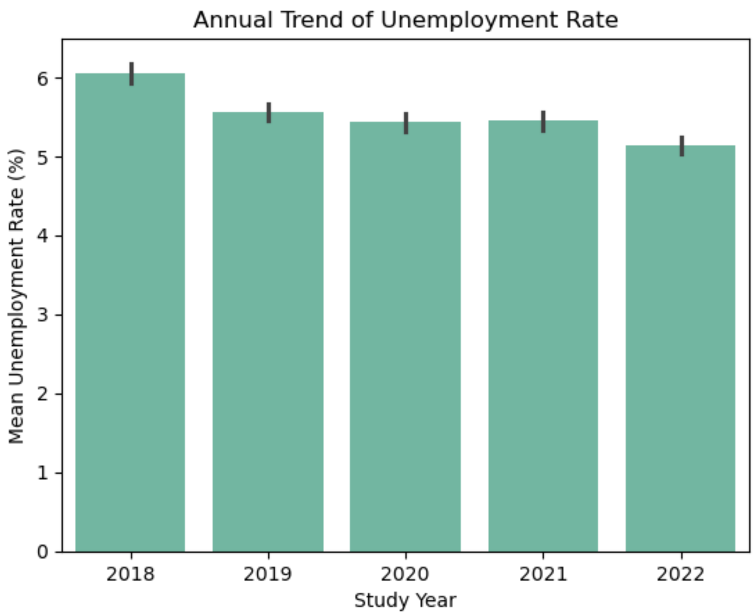

Figure 11: *Bar chart showing the annual trend of unemployment rate in the United States of America*

Comparing each gender, the pie chart in Figure 12 indicates that unemployment rates are slightly higher among males compared to females in the United States, accounting for 52.26% and 47.74% respectively. The pie chart visualisation method was chosen because it was most effective for showing part-to-whole relationships when the number of categories is limited. Pie Chart improves interpretability and reduces visual complexity (Few, 2012).

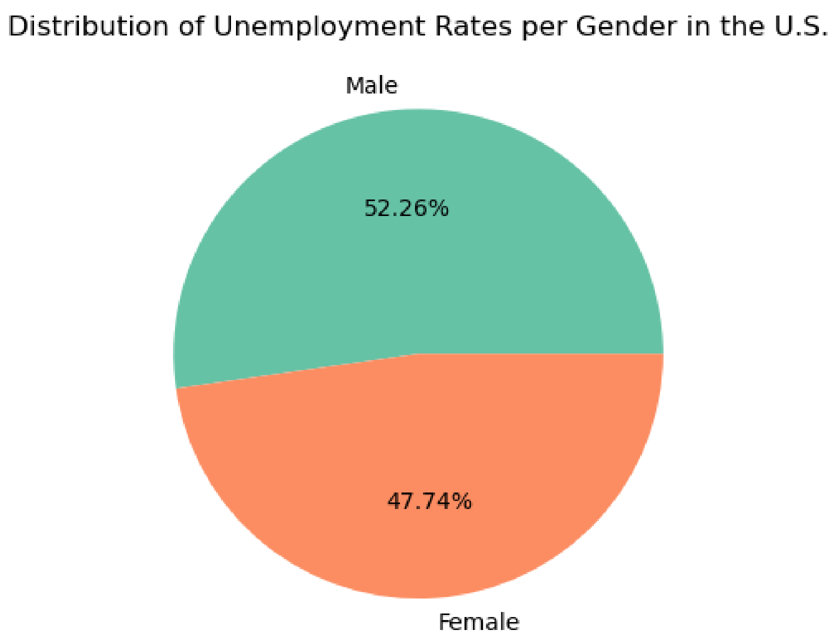

Figure 12: *Pie chart comparing the unemployment rates of both gender across five (5) years*

3.5 Analysing the Relationships Between Other Variables with High Correlation Scores

To enable the visualisation of relationships between two or more variables, a reusable function was developed to streamline the creation of scatter plots and automate annotation processes, thereby reducing repetitive code within the notebook and improving workflow efficiency. The function scales the dataset variables using normalisation method before visualising to ensure that the scales of the x and y variables are the same to eliminate any skewness of the chart due to varying variables scales.

Scatter plot was preferred for visualising correlation between two variables or more because they are particularly effective for identifying and interpreting correlations between continuous variables, allowing individual data points to be clearly displayed and patterns such as positive, negative, or non-linear relationships to be easily observed (Tufte, 2001).

```
def visualiseReationship(x, y, xlabel, ylabel, data_label = None, max_annot_text = "", marker = ".", transparent = False, size = 20):
    #scale the variables around their mean
    x_norm = (x - x.min()) / (x.max() - x.min()) #normalise the values of x
    y_norm = (y - y.min()) / (y.max() - y.min()) #normalise the values of y
    opacity = 1 if transparent == False else 0.6

    plt.scatter(x_norm, y_norm, marker=marker, label=data_label, alpha=opacity, s=size)

    if max_annot_text:
        plt.annotate(max_annot_text, xy = (x_norm.max()+0.03, y_norm.max()-0.01), bbox=dict(boxstyle="round", fc=(1.0, 0.7, 0.7), ec="none"))
    plt.xlabel(f"{xlabel} (Normalised)")
    plt.ylabel(f"{ylabel} (Normalised)")
```
✓ 0.0s

Figure 13: *A reusable function for plotting scatter plot and making annotation on it. For showing relationship between 2 or 3 variables*

A similar reusable function was created for plotting Seaborn 2D Kernel Density Estimation plot as a complementary technique because they extend beyond individual data points to reveal the underlying distribution density of observations. This makes it easier to identify areas of concentration, hidden structure, and multivariate distribution patterns that may not be immediately visible in scatter plots alone.

The following subsections describe the bivariate analysis of the variables that showed positive correlations and the outcomes.

3.5.1 Visualising the Relationship between Centre-Based Care Prices and Family Child-Care Prices for each State County

The strong positive correlation of 90% observed between Centre-Based Care Prices and Family Child-Care Prices is further illustrated in the figure below, which demonstrates a clear linear relationship between the two variables. This suggests that increases in centre-based care costs are generally associated with corresponding increases in family child-care prices, indicating that both forms of childcare pricing tend to move in tandem across the dataset.

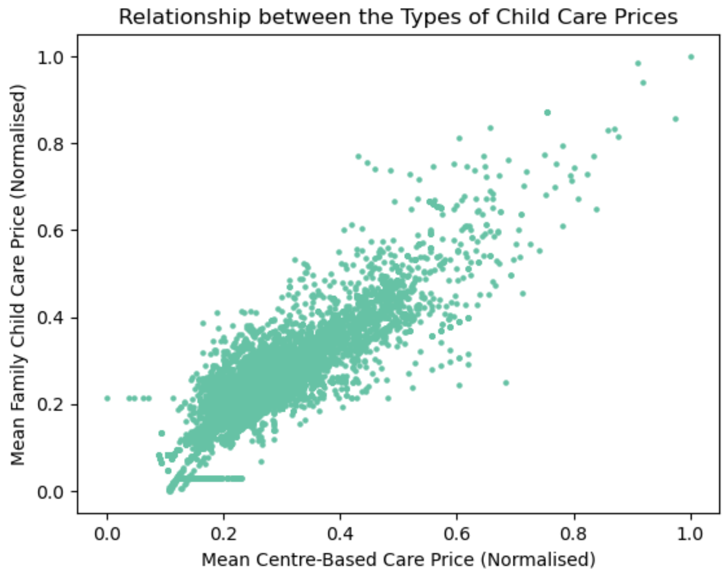

Figure 14: *Scatter plot showing the Correlation Between Ccentre-Based Care Price and Family Child Care Prices*

3.5.2 Visualising the Correlation between the 2022 Median Household Income and Child-Care Price

Using the Seaborn 2D kernel density estimate (KDE) plot shown in Figure 15, the relationship between 2022 Median Household Income and Child-Care Prices displays a weak but noticeable directional pattern, suggesting a slight positive correlation between the two variables. However, the majority of data points are densely concentrated in the lower-left region of the plot, indicating that most counties tend to have both lower household incomes and lower childcare costs. This clustering highlights the uneven distribution of economic conditions across the dataset, where only a smaller proportion of observations extend into higher income and higher childcare price ranges.

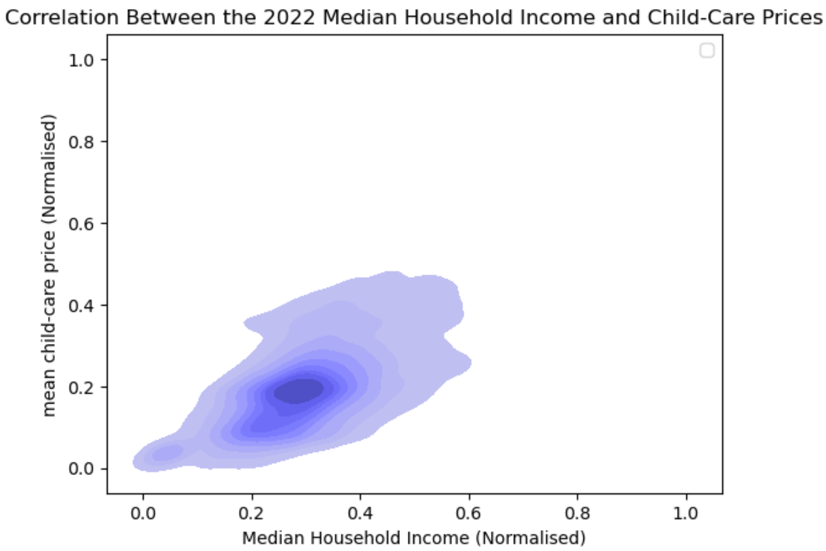

Figure 15: *A 2D KDE plot showing the correlation between mdian household income and average child-care prices*

3.5.3 Visualising the relationship between Poverty Population and Average Child-Care Prices in each State County

As shown in the scatter plot shown in Figure 16, the plot suggests that most state counties have relatively low poverty population values, with only a few notable outliers present in the upper range. The relationship between poverty population and childcare prices does not appear to follow a clear linear pattern, as childcare costs vary considerably across counties regardless of poverty population size. This wide dispersion suggests that other socioeconomic factors may be influencing childcare pricing, resulting in a weak or non-existent direct linear association between the two variables.

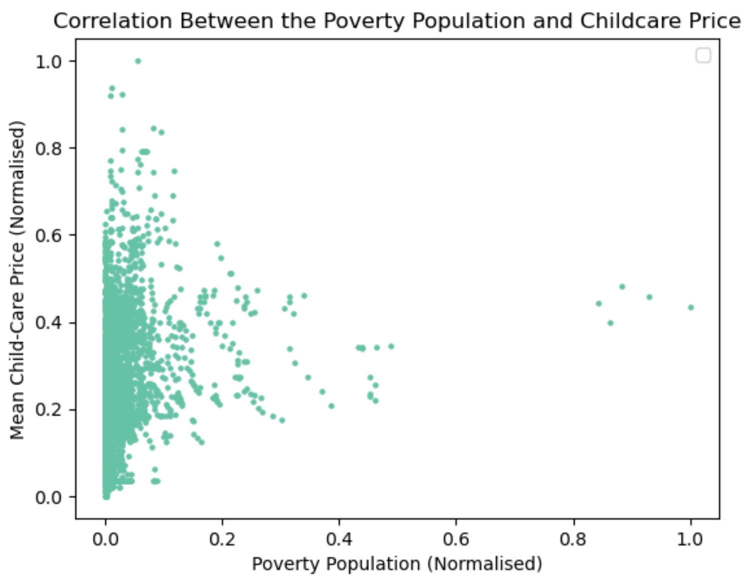

Figure 16: *Scatter Plot showing relationship between poverty population count and child-care cost*

4.0 In-Depth Multivariate Data Analysis

The primary objectives of this section are to:

1. Analyse the trend of average U.S unemployment rate by gender against the annual poverty rate across the years from 2018 to 2022.
2. Analyse the relationship between unemployment rate and poverty rate per gender per state.
3. Create choropleth visualisation of median household income and poverty rate across states.
4. Analyse the effects of proportion of households with only one working parent on the poverty rate in the states.
5. Analyse the relationship between average weekly child-care costs and poverty rates across states.

4.1 Analysing the Trend of Average U.S Unemployment Rate by Gender Against the Annual Poverty Rate Across the Years

There was a consistent downward trend in the average unemployment rate in the United States for both males and females between 2018 and 2022, with a year-on-year reduction of approximately -0.23%. A similar pattern was observed in the poverty rate, which declined at a comparable annual rate of around -0.39%. The lowest values were recorded in 2022, where the poverty rate stood at 12.85%, male unemployment at 5.3%, and female unemployment at 4.9%.

The combined visualisation presented below highlights these trends and suggests a potential relationship between unemployment and poverty over time, as both indicators exhibit a broadly similar downward trajectory. A hybrid chart combining grouped bar charts for gender-based unemployment rates and a line chart for poverty rate was used because it allows multiple related variables to be compared within a single coherent visual framework over the same time period.

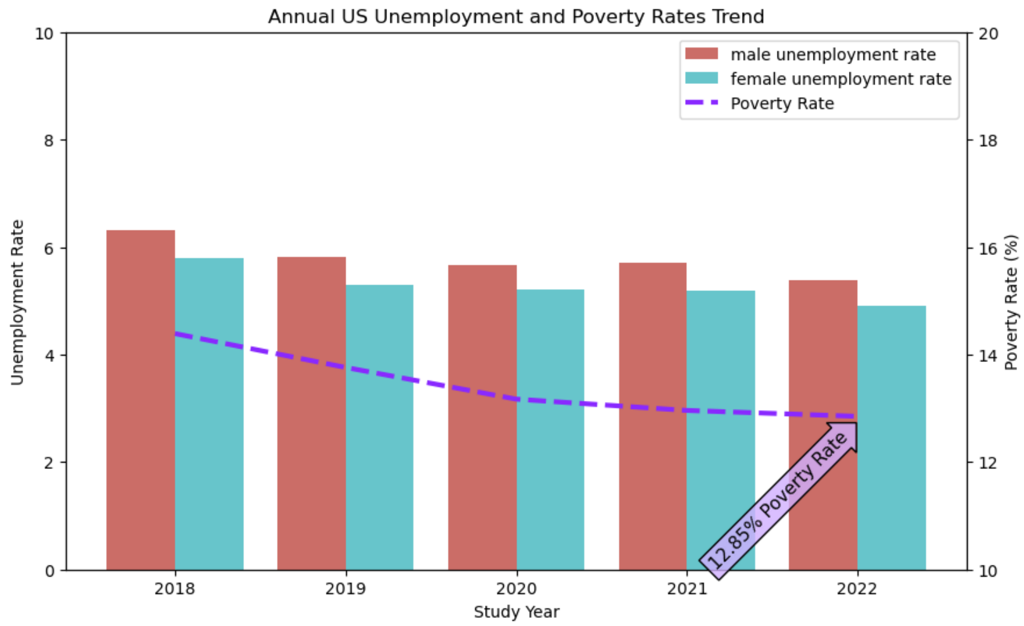

Figure 17: *A combined visualisation showing the decline of annual trend of unemployment rate and poverty rate*

4.2 Analysing the Relationship Between Unemployment Rate and Poverty Rate per Gender per State

At the state level, the three-variable scatter plot in Figure 18 further illustrates the distribution of observations and indicates a broadly positive relationship between unemployment rates and poverty rates across states, disaggregated by gender. The visualisation also highlights Puerto Rico as a notable outlier, recording the highest levels of both unemployment and poverty in the United States at 15.81% and 43.38% respectively.

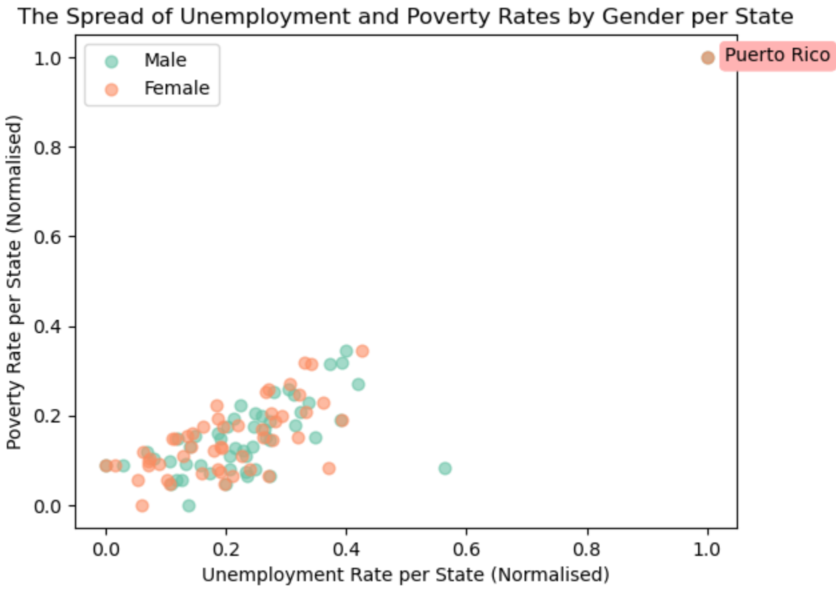

Figure 18: *Relationship between unemployment rate and poverty rate per state per gender*

4.3 Choropleth Visualisation of Median Household Income and Poverty Rate Across States

The interactive choropleth visualisation was implemented using Plotly due to its flexibility in handling interactive geographic plots and its strong support for custom map layers. Unlike static plotting libraries, Plotly enables dynamic exploration of spatial data through hover information, zooming, and clear colour scaling, which is particularly useful for comparing socioeconomic indicators across states.

To extend the default geographic coverage provided by Plotly Express, a custom publicly available U.S States GeoJSON file was sourced from the PublicaMundi GitHub (PublicaMundi, 2014). This dataset was merged with the main state-level dataset to ensure accurate spatial mapping of all regions, including Puerto Rico, which is not included in Plotly’s default United States choropleth configuration. This adjustment was necessary to maintain completeness and avoid exclusion of key territorial data in the analysis

In addition, geographic centroid coordinates for each state were incorporated from the Google Public Data repository (Google, 2024). These centroids were integrated into the dataset to support more precise spatial alignment and to enhance the representation of poverty rates using location-based anchoring.

As shown in Figure 19, the following insights and conclusions were derived from the choropleth visualisation;

- The combined choropleth and bubble map visualisation illustrates the distribution of median household income across U.S. states.
- From the visualisation, it can be observed that many southern states in the United States tend to have lower median household incomes compared with other regions. The same pattern is reflected in poverty rates, with southern states generally exhibiting higher levels of poverty relative to other parts of the country.
- The visualisation highlights a significant outlier in the southern region, namely Puerto Rico, which records the highest poverty rate at 45.25% and the lowest median household income in the dataset at $22,634.71.

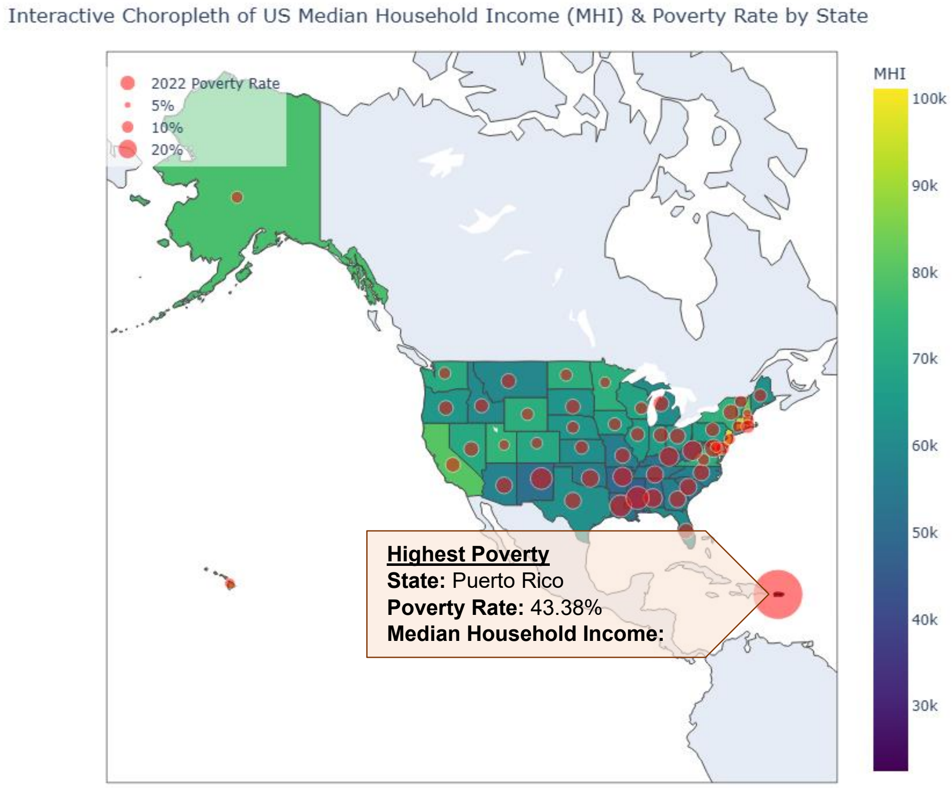

Figure 19: Choropleth of the Median Household Income and Poverty Rate per State

- States with the highest median household incomes are predominantly located in the eastern coastal regions of the country.

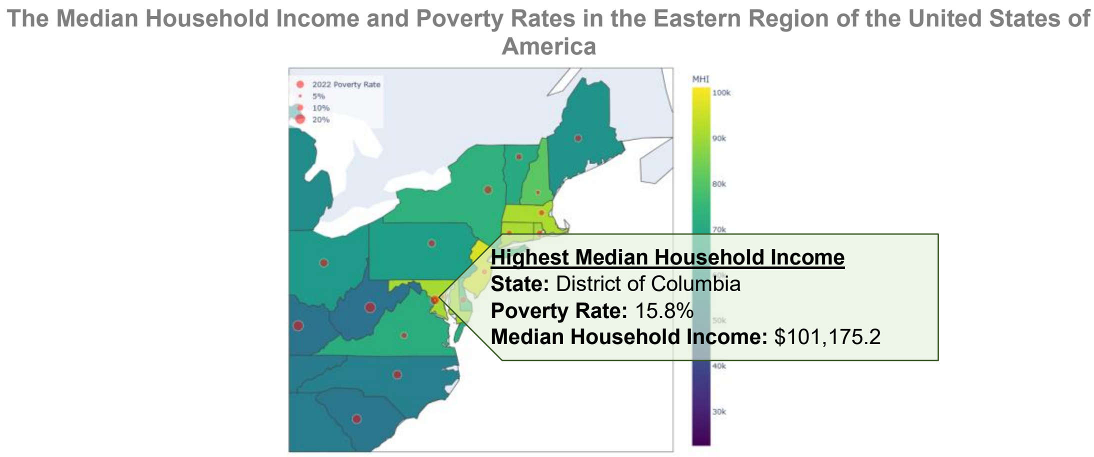

Figure 20: Choropleth visualisation showing the median household incomes and poverty rates in the eastern region of the United States

4.4 Analysing the Effects of Proportion of Households with only one Working Parent on the Poverty Rate in the States

The combined lollipop and line chart visualisation above indicates that poverty rates tend to increase alongside the proportion of one-breadwinner households across states. This pattern is particularly evident in Puerto Rico, where more than 70% of households with children under the age of 18 rely on a single working parent and where the highest poverty rate in the dataset is also observed. The visualisation suggests that states with a greater dependence on a single income source may be more vulnerable to financial hardship, as household earnings are less resilient to economic shocks and rising living costs. Although this relationship does not necessarily imply causation, the overall trend points to a meaningful association between household employment structure and poverty levels.

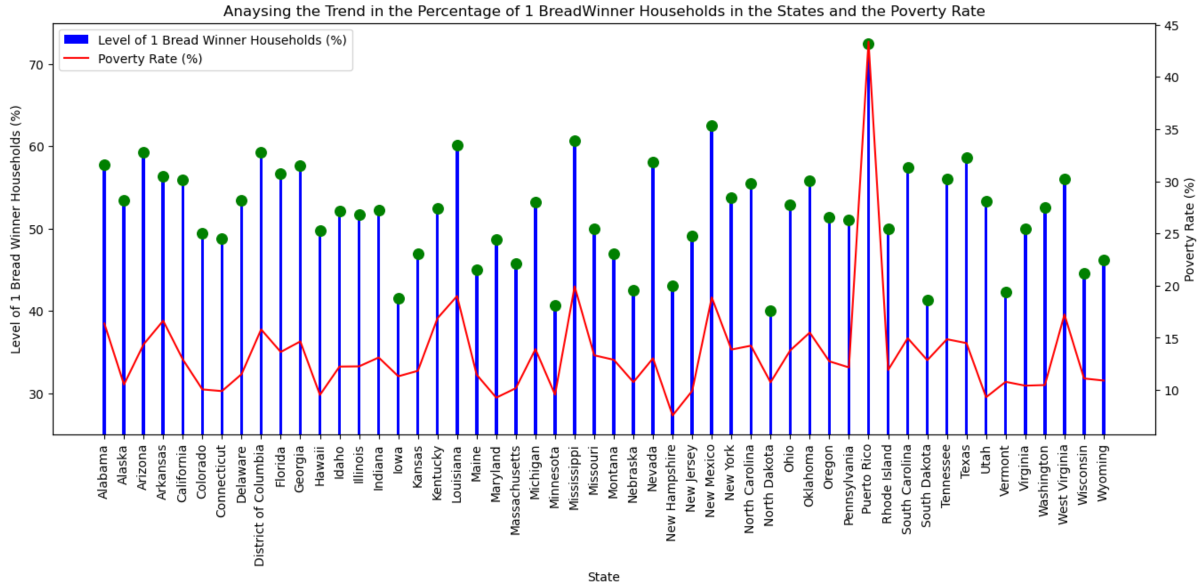

Figure 21: *Combined lollipop chart showing the proportion of households with only one provider for each state together with a line chart showing the poverty rate across each State*

A lollipop chart was selected because it offers a less cluttered alternative to a conventional bar chart while preserving the ability to make accurate comparisons between categories. By reducing the amount of non-data ink displayed, lollipop charts place greater emphasis on the data values themselves, thereby improving visual efficiency and readability (Tufte, 2001). The combination of a lollipop chart and a line chart also facilitate comparison between two related variables, enabling patterns and trends to be identified more easily across States.

4.5 Analysing the Relationship Between Average Weekly Child-Care Costs and Poverty Rates Across States

The combined lollipop and line chart reveals a weak inverse relationship between average weekly child-care costs and poverty rates across U.S. states. While the overall trend suggests that states with higher childcare costs tend to exhibit slightly lower poverty rates, the relationship is not particularly strong and displays considerable variation between states.

The visualisation highlights the District of Columbia as the jurisdiction with the highest average weekly childcare cost, recorded at $379.03. Despite these elevated childcare expenses, its poverty rate stands at 15.8%, which is broadly aligned with the national average. In contrast, Puerto Rico records the lowest average weekly childcare cost at

$75.03 but simultaneously has the highest poverty rate in the dataset. These findings suggest that childcare costs alone are unlikely to be a primary determinant of poverty levels and that broader socioeconomic factors, such as employment opportunities, household income, and economic development, may exert a greater influence on poverty outcomes.

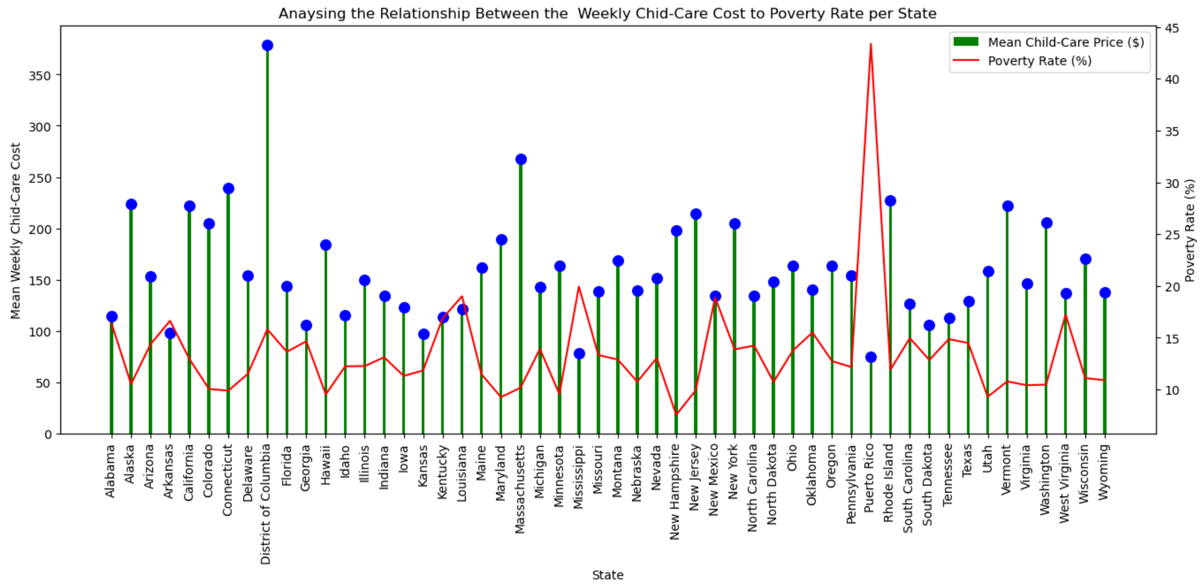

Figure 22: *A combined lollipop chart and line chart showing average child-care prices and poverty rates respectively with no visible correlation*

5.0 Summary of Analysis Insights and Conclusion

This project explored the underlying factors associated with poverty rates across the United States using data from the National Database of Childcare Prices between 2018 and 2022. Through a combination of data preparation, exploratory analysis, statistical visualisation, and geographic mapping techniques, the study examined how poverty levels vary in relation to unemployment, household income, childcare costs, poverty population size, and household employment structures.

The findings indicate that unemployment rates and poverty rates share a positive relationship, with states exhibiting higher unemployment generally recording higher levels of poverty. Analysis of annual trends further revealed that both indicators declined steadily between 2018 and 2022, suggesting that improvements in labour market conditions were accompanied by reductions in poverty. Median household income was also found to be an important factor, with states reporting lower household incomes generally experiencing higher poverty rates. Geographic visualisations reinforced this observation, highlighting regional disparities across the country, particularly within southern states.

The study also identified household employment structure as a notable contributor to poverty outcomes. States with a higher proportion of one-breadwinner households tended to record higher poverty rates, suggesting that reliance on a single source of income may increase economic vulnerability. In contrast, the relationship between childcare costs and poverty rates appeared relatively weak. While childcare prices varied substantially across states, the analysis showed that low childcare costs did not necessarily correspond to lower poverty levels, indicating that childcare affordability alone does not adequately explain variations in poverty.

A significant finding throughout the analysis was the prominence of Puerto Rico as an outlier. The territory consistently recorded the highest poverty rates, among the highest unemployment rates, and the lowest median household income levels. These characteristics distinguished Puerto Rico from other states and highlighted the substantial socioeconomic disparities that exist within the broader United States.

Overall, the visualisations demonstrated that poverty is influenced by a combination of interconnected socioeconomic factors rather than a single variable. The use of heatmaps, scatter plots, kernel density plots, choropleth maps, and combined comparative charts provided complementary perspectives that enabled both statistical relationships and geographic patterns to be identified effectively.

6.0 Recommendations

1. Expanding government-led employment programmes and economic development initiatives could further strengthen socioeconomic conditions across many regions of the United States and contribute to continued reductions in poverty levels.
2. Improvements to public resource allocation and regional development policies may promote a more balanced distribution of economic opportunities across states, particularly within the Southern region and, most notably, Puerto Rico, where poverty levels remain substantially higher than the national average.
3. Educational and public awareness initiatives that highlight the economic advantages of dual-income households may help reduce financial vulnerability in areas where a large proportion of families rely on a single income earner.

7.0 Limitations and Future Considerations

While this study provides useful insights into the factors associated with poverty across U.S. states, several limitations should be acknowledged. Firstly, the analysis is largely observational and therefore cannot establish causality between variables such as unemployment, childcare costs, and poverty rates. The relationships identified in the visualisations show association rather than direct cause-and-effect.

In addition, the dataset focuses on selected socioeconomic indicators and does not include other potentially influential factors such as education levels, housing costs, healthcare access, or inflation rates. Future work could address these limitations by incorporating additional socioeconomic datasets to build a more comprehensive model of poverty determinants.

Extending the analysis by conducting deeper county-level analysis may provide more detailed regional insights and improve the robustness of findings.

8.0 Ethical Considerations

This study used publicly available and anonymised socioeconomic datasets. No personal or identifiable information was accessed or processed and as a result, the risk to individual privacy is minimal.

Care was taken to ensure that data handling, cleaning, and imputation processes did not distort findings in a way that could misrepresent specific regions or populations. All visualisations and interpretations were presented objectively to avoid misleading conclusions or reinforcing stereotypes about particular States or communities.

9.0 References

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10.0 The Data Visualisation Poster

This poster presents a summary of the in-depth exploratory data analysis of poverty across U.S. States. The analysis investigates how key factors such as unemployment rates, median household income, childcare costs, and household structure relate to variations in poverty levels.

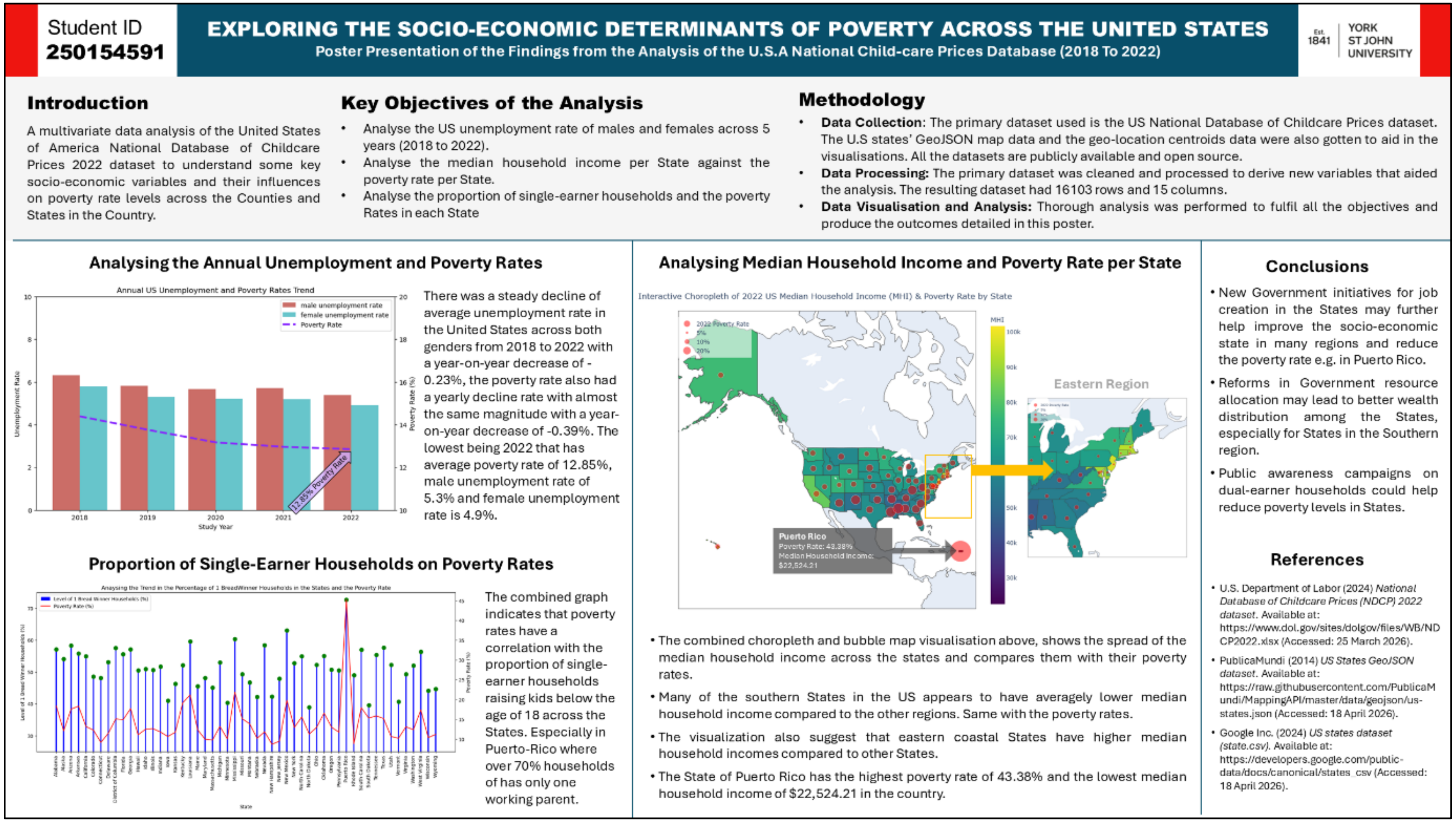


***Figure 23:** The poster showing summary of the analysis project and key details*

11.0 Appendix

The Complete Jupyter Python Code Notebook

Exploring the Socioeconomic Determinants of Poverty Across the United States from 2018 to 2022: A Data Analysis and Visualisation Project

1.0) Overview

An analysis of the National Database of Childcare Prices 2022 dataset to understand the key socio-economic variables across the States in the United States of America and their influence on poverty rate levels in the Country.

2.0) Data Import and Preparation

2.1) Importing the Data and Python Packages Necessary

Data Source
The National Database of Childcare Prices (NDCP) under the United States Department of Labour. https://dataportal.dol.gov/datasets/10278

```python
#import necessary libraries and packages
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from pathlib import Path
import warnings
```

[468] ✓ 0.0s Python

```python
warnings.filterwarnings("ignore") #to avoid clogging the notebook with warnings, they will be filtered
```

[469] ✓ 0.0s Python

```python
#import the raw data
raw_df = pd.read_csv("./dataset/NDCP_2022_dataset.csv")
raw_df.info()
```

[470] ✓ 3.5s Python

```
<class 'pandas.DataFrame'>
RangeIndex: 48308 entries, 0 to 48307
Columns: 370 entries, state_name to ifemp_p_state
dtypes: float64(200), int64(167), str(3)
memory usage: 136.4 MB
```

```python
raw_df.describe().T
```

[471] ✓ 1.5s Python

	count	mean	std	min	25%	50%	75%	max
county_fips_code	48308.0	31394.875362	16291.031838	1001.0	19032.5	30025.0	46107.0	72153.0
studyyear	48308.0	2014.999959	4.320660	2008.0	2011.0	2015.0	2019.0	2022.0
emr_16	48308.0	54.860845	8.743072	11.0	49.5	55.6	61.2	98.3
femr_16	48308.0	50.968001	7.994420	13.6	46.0	51.4	56.5	100.0
memr_16	48308.0	58.979668	10.430401	8.0	52.9	60.2	66.4	100.0
...	...	...	...	...	...	...	...	...
imemp_n_state	48308.0	1.000000	0.000000	1.0	1.0	1.0	1.0	1.0
ifemp_n_state	48308.0	1.000000	0.000000	1.0	1.0	1.0	1.0	1.0
iemp_p_state	48308.0	1.000000	0.000000	1.0	1.0	1.0	1.0	1.0
imemp_p_state	48308.0	1.000000	0.000000	1.0	1.0	1.0	1.0	1.0
ifemp_p_state	48308.0	1.000000	0.000000	1.0	1.0	1.0	1.0	1.0

367 rows × 8 columns

```
raw_df.head()
```

[472] 0.1s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	emr_16	femr_16	memr_16	emr_20to64	femr_20to64	...	ifemp_service_state	iemp_sales_st
0	Alabama	AL	Autauga County	1001	2008	61.5	56.1	67.3	71.4	66.3	...	1	
1	Alabama	AL	Autauga County	1001	2009	60.6	54.8	67.0	72.5	66.9	...	1	
2	Alabama	AL	Autauga County	1001	2010	60.4	54.9	66.4	72.3	67.4	...	1	
3	Alabama	AL	Autauga County	1001	2011	58.7	52.4	65.6	71.0	64.9	...	1	
4	Alabama	AL	Autauga County	1001	2012	57.7	52.2	63.8	70.7	65.1	...	1	

5 rows × 370 columns

2.2) Processing and Preparing the Dataset for Analysis

The dataset contains 48,308 observations and 370 attributes. Half of the attributes are metadata attributes that specify whether data was systematically imputed for a particular missing attribute of an observation e.g. `iemr_16_state` is a binary data that specifies whether the 'employment rate of females in the county' for an observation was systematically imputed. These metadata columns and other attributes with low level of relevance to the key objectives of analysis will be removed.

Since the analysis focuses on the period of 2018 to 2022, only the observations captured at that period will be retained.

The relevant attributes are:

Field Name	Field Description	Intended Data Type
state_name	Name of state for which the data are being entered.	categorical
state_abbreviation	Two-letter state abbreviation	categorical
county_name	Name of the county for which data are being entered.	categorical
county_fips_code	Five-digit number that uniquely identifies the county in a state.	categorical
studyyear	Year in which data were collected for the market rate survey.	categorical
unr_16	Unemployment rate of the population aged 16 years old or older	numeric
funr_16	Unemployment rate of the female population aged 16 years old or older	numeric
munr_16	Unemployment rate of the male population aged 16 years old or older	numeric
mhi_2022	Median household income expressed in 2022 dollars	numeric
totalpop	Count of the total population	numeric
h_under6_fwork	Number of households with children under 6 years old with only the father working	numeric
h_under6_mwork	Number of households with children under 6 years old with only the mother working	numeric
h_under6_singlem	Number of households with children under 6 years old with a single mother	numeric
h_6to17_fwork	Number of households with children between 6 and 17 years old with only the father working	numeric
h_6to17_mwork	Number of households with children between 6 and 17 years old with only the mother working	numeric
h_6to17_singlem	Number of households with children between 6 and 17 years old with a single mother	numeric
h_under6_bothwork	Number of households with children under 6 years old with two working parents	numeric
h_6to17_bothwork	Number of households with children between 6 and 17 years old with two working parents	numeric
mcbto5	Weekly, full-time median price charged for Center-Based Care for those aged 0 to 5 months	numeric
mc6to11	Weekly, full-time median price charged for Center-Based Care for those aged 6 to 11 months	numeric
mc12to17	Weekly, full-time median price charged for Center-Based Care for those aged 12 to 17 months	numeric
mc18to23	Weekly, full-time median price charged for Center-Based Care for those aged 18 to 23 months	numeric
mc24to29	Weekly, full-time median price charged for Center-Based Care for those aged 24 to 29 months	numeric
mc30to35	Weekly, full-time median price charged for Center-Based Care for those aged 30 to 35 months	numeric
mc36to41	Weekly, full-time median price charged for Center-Based Care for those aged 36 to 41 months	numeric
mc42to47	Weekly, full-time median price charged for Center-Based Care for those aged 42 to 47 months	numeric
mc48to53	Weekly, full-time median price charged for Center-Based Care for those aged 48 to 53 months	numeric
mc54tosa	Weekly, full-time median price charged for Center-Based Care for those aged 54 months to school age	numeric

mcsa	Weekly, full-time median price charged for Center-Based Care for those who are school age	numeric
mfccbto5	Weekly, full-time median price charged for Family Childcare for those aged 0 to 5 months	numeric
mfcc6to11	Weekly, full-time median price charged for Family Childcare for those aged 6 to 11 months	numeric
mfcc12to17	Weekly, full-time median price charged for Family Childcare for those aged 12 to 17 months	numeric
mfcc18to23	Weekly, full-time median price charged for Family Childcare for those aged 18 to 23 months	numeric
mfcc24to29	Weekly, full-time median price charged for Family Childcare for those aged 24 to 29 months	numeric
mfcc30to35	Weekly, full-time median price charged for Family Childcare for those aged 30 to 35 months	numeric
mfcc36to41	Weekly, full-time median price charged for Family Childcare for those aged 36 to 41 months	numeric
mfcc42to47	Weekly, full-time median price charged for Family Childcare for those aged 42 to 47 months	numeric
mfcc48to53	Weekly, full-time median price charged for Family Childcare for those aged 48 to 53 months	numeric
mfcc54tosa	Weekly, full-time median price charged for Family Childcare for those aged 54 months to school age	numeric
mfccsa	Weekly, full-time median price charged for Family Childcare for those who are school age	numeric
pr_p	Poverty rate for individuals	numeric

i) Filtering for the Relevant Columns and rows

```
#filter for the relevant variables
filtered_df = raw_df[["state_name", "state_abbreviation","county_name", "county_fips_code", "studyyear", "unr_16", "funr_16", "munr_16", "mhi_2022", "totalpo
filtered_df = filtered_df[filtered_df["studyyear"] >= 2018] #filter only data from 2018 to 2022
filtered_df = filtered_df.reset_index(drop=True) #reset the indexes so that the dataset is arranged well
filtered_df.head()
```

[473] 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	...	mfcc12to17	mfcc18to23	mfcc24to29
0	Alabama	AL	Autauga County	1001	2018	4.2	3.4	4.9	69367	55200	...	112.33	112.33	111.58
1	Alabama	AL	Autauga County	1001	2019	3.7	3.2	4.1	68128	55380	...	117.55	117.55	116.05
2	Alabama	AL	Autauga County	1001	2020	2.9	2.4	3.3	66679	55639	...	122.78	122.78	120.53
3	Alabama	AL	Autauga County	1001	2021	2.8	2.3	3.3	68299	58239	...	128.00	128.00	125.00
4	Alabama	AL	Autauga County	1001	2022	2.8	2.6	2.9	68315	58761	...	133.23	133.23	129.48

5 rows × 41 columns

```
filtered_df.shape #show the shape of the dataset
```

[474] 0.0s Python

```
(16103, 41)
```

By focusing only on variables and rows that are relevant to this analysis project, the total number of relevant rows are 16,103 and the number of variables is 40.

ii) Treating Blanks

Checking the rows for blanks

```
print("List of columns and their number of blanks rows:")
filtered_df.isna().sum().sort_values(ascending=False)
```

[475] 0.0s Python

```
List of columns and their number of blanks rows:

mfccsa                3140
mfcc24to29            3026
mfcc30to35            3026
mfcc6to11             3019
mfccbto5              3019
mfcc18to23            3018
mfcc12to17            3018
mfcc36to41            3014
mfcc48to53            3014
mfcc42to47            3014
mfcc54tosa            3014
mcsa                  2673
mc6to11               2639
mcbto5                2639
mc18to23              2627
mc12to17              2627
mc30to35              2618
mc24to29              2618
mc54tosa              2602
mc48to53              2602
mc36to41              2602
mc42to47              2602
state_name               0
county_name              0
state_abbreviation       0
...
h_6to17_fwork            0
h_6to17_mwork            0
h_6to17_singlem          0
pr_p                     0
dtype: int64
```

The blank rows of each attribute for each State will be replaced by the mean of that attribute for that State alone instead of the general attribute's column to avoid filling the blanks of some State with values that may be larger than average due to some states with high variable values.

```
states_abvs = filtered_df["state_abbreviation"].value_counts().index #getting the list of states' abbreviations
```

[476] 0.0s Python

```
#creating a reusable function for filling all blank rows in all the columns of a dataframe with their respective means
def fill_blanks_with_mean(series):
    if(series.dtype != "string"): #this is to skip string-type columns to avoid having error when the mean is being calculated
        n_series = series.fillna(series.mean())
        return n_series
    else:
        return series
```

[477] 0.0s Python

```
#creating a reusable function for filling all blank rows in all the columns of a dataframe with their respective medians
def fill_blanks_with_median(series):
    if(series.dtype != "string"): #this is to skip string-type columns to avoid having error when the median is being calculated
        n_series = series.fillna(series.median())
        return n_series
    else:
        return series
```

[478] 0.0s Python

```
#replacing blanks in each subset of the data for each State with their respective averages
for state_ID in states_abvs:
    filtered_df[filtered_df["state_abbreviation"] == state_ID] = filtered_df[filtered_df["state_abbreviation"] == state_ID].apply(fill_blanks_with_mean)
```

[479] 1.5s Python

```python
def show_cols_with_blanks(df):
    count = 0
    for col in filtered_df.columns:
        blank_count = df[col].isna().sum()
        if(blank_count > 0):
            count += 1
            print(f"'{col}' contains {blank_count} blanks")

    if(count == 0):
        print("There are no more blank rows")
```
[480] ✓ 0.0s Python

```python
show_cols_with_blanks(filtered_df) #confirming if any column still contain blanks, if there is any, show them
```
[481] ✓ 0.0s Python

```
'mcbto5' contains 625 blanks
'mc6to11' contains 625 blanks
'mc12to17' contains 625 blanks
'mc18to23' contains 625 blanks
'mc24to29' contains 625 blanks
'mc30to35' contains 625 blanks
'mc36to41' contains 625 blanks
'mc42to47' contains 625 blanks
'mc48to53' contains 625 blanks
'mc54tosa' contains 625 blanks
'mcsa' contains 630 blanks
'mfccbto5' contains 625 blanks
'mfcc6to11' contains 625 blanks
'mfcc12to17' contains 625 blanks
'mfcc18to23' contains 625 blanks
'mfcc24to29' contains 625 blanks
'mfcc30to35' contains 625 blanks
'mfcc36to41' contains 625 blanks
'mfcc42to47' contains 625 blanks
'mfcc48to53' contains 625 blanks
'mfcc54tosa' contains 625 blanks
'mfccsa' contains 630 blanks
```

Checking the names of the States that still have blank rows

```python
print("The names of states still having blanks:")
filtered_df[filtered_df["mc18to23"].isna()]["state_name"].value_counts() #check the names of the States with blanks
```
[482] ✓ 0.0s Python

```
The names of states still having blanks:

state_name
Indiana       460
New Mexico    165
Name: count, dtype: int64
```

The States of Indiana and New Mexico appears to not have any values inputed for all categories of **"child care prices"**, these will be filled with the national **median** from all the States. The median is used instead of the mean to avoid influencing the values with high variables/outliers from some States.

```python
filtered_df = filtered_df.apply(fill_blanks_with_median) #fill all remaining blank rows in each attribute with the national median of each attribute
```
[483] ✓ 0.0s Python

```python
show_cols_with_blanks(filtered_df) #check if there is still any blank rows
```
[484] ✓ 0.0s Python

```
There are no more blank rows
```

```python
#checking the observations from the State of Indiana again
filtered_df[filtered_df["state_abbreviation"] == "IN"].head()
```
[485] ✓ 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	...	mfcc12to17	mfcc18to23	mfcc24to2
3488	Indiana	IN	Adams County	18001	2018	3.6	3.7	3.6	61021	35195	...	130.943333	129.25	124.76
3489	Indiana	IN	Adams County	18001	2019	3.6	3.7	3.5	60905	35376	...	130.943333	129.25	124.76
3490	Indiana	IN	Adams County	18001	2020	3.0	3.1	3.0	60619	35544	...	130.943333	129.25	124.76

3491	Indiana	IN	Adams County	18001	2021	3.0	3.3	2.8	61080	35685	...	130.943333	129.25	124.76
3492	Indiana	IN	Adams County	18001	2022	3.2	3.0	3.4	61731	35827	...	130.943333	129.25	124.76

5 rows × 41 columns

iii) Feature Engineering

New variables will be generated by merging some related variables together as they will be analysed as a whole, for example, the attributes representing **Number of houseolds where both parents are working** were recorded in child age-group clusters, hence the values in each age-group type will be merged into one whole variable.

A variable for **poverty population** will be calculated as this will enable the accurate calculation of unbiased state-level or national-level poverty rates since the population in each county is important to be put into consideration because 50% poverty rate in a county with 200 people is not the same weight as 50% poverty rate in a county with 10,000 people.

The dataset will be generally normalised into the first normal form and redundant attributes will be dropped, for exsample, **'unemployment rate'** where the value can be derived from the gender-based unemployment rate attributes.

Data manipulation steps:

- Merging the various attributes that represent the number of households with children under 18 and only one parent working into a single attribute.

```python
filtered_df["1_breadwinner_hh"] = filtered_df[["h_under6_fwork", "h_under6_mwork", "h_under6_singlem", "h_6to17_fwork", "h_6to17_mwork", "h_6to17_singlem"]]
#Dropping the extra columns that are no more needed
filtered_df = filtered_df.drop(["h_under6_fwork", "h_under6_mwork", "h_under6_singlem", "h_6to17_fwork", "h_6to17_mwork", "h_6to17_singlem"], axis=1)
filtered_df.head(1)
```

[486] ✓ 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	...	mfcc18to23	mfcc24to29	mfcc30to35
0	Alabama	AL	Autauga County	1001	2018	4.2	3.4	4.9	69367	55200	...	112.33	111.58	110.93

- Merging the two attributes that represent number of households with children under 18 that has both parents working

```python
filtered_df["2_breadwinners_hh"] = filtered_df[["h_under6_bothwork", "h_6to17_bothwork"]].sum(axis=1).astype("int64")
filtered_df = filtered_df.drop(["h_under6_bothwork", "h_6to17_bothwork"], axis=1) #dropping the extra columns that are no more needed
filtered_df.head(1)
```

[487] ✓ 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	...	mfcc24to29	mfcc30to35	mfcc36to41
0	Alabama	AL	Autauga County	1001	2018	4.2	3.4	4.9	69367	55200	...	111.58	110.93	110.93

1 rows × 35 columns

- Merging the values of the median centre-base care prices accross the children age groups into a single attribute per county using the mean of them all

```python
# find the mean accross all the age-groups of the median centre-based care prices and assign it into a new single attribute
filtered_df["avg_centrebased_care_price"] = filtered_df[["mcbto5", "mc6to11", "mc12to17", "mc18to23", "mc24to29", "mc30to35", "mc36to41", "mc42to47", "mc48t
#dropping the extra columns that are no more needed
filtered_df = filtered_df.drop(["mcbto5", "mc6to11", "mc12to17", "mc18to23", "mc24to29", "mc30to35", "mc36to41", "mc42to47", "mc48to53", "mc54tosa", "mcsa"]
filtered_df.head(1)
```

[488] ✓ 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	...	mfcc30to35	mfcc36to41	mfcc42to47
0	Alabama	AL	Autauga County	1001	2018	4.2	3.4	4.9	69367	55200	...	110.93	110.93	110.93

1 rows × 25 columns

- Using the mean, the median family childcare prices accross the age-groups will be merged into one for each county

```
filtered_df["avg_family_childcare_price"] = filtered_df[["mfccbto5", "mfcc6to11", "mfcc12to17", "mfcc18to23", "mfcc24to29", "mfcc30to35", "mfcc36to41", "mfc
#dropping the extra columns that are no more needed
filtered_df = filtered_df.drop(["mfccbto5", "mfcc6to11", "mfcc12to17", "mfcc18to23", "mfcc24to29", "mfcc30to35", "mfcc36to41", "mfcc42to47", "mfcc48to53", "r
filtered_df.head(1)
```

[489] 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	unr_16	funr_16	munr_16	mhi_2022	totalpop	pr_p	1_breadwinner_hh	2_breadwinners_hh
0	Alabama	AL	Autauga County	1001	2018	4.2	3.4	4.9	69367	55200	15.4	6127	5523

- Calculating the poverty population per County

```
#multiply the total population by poverty rate in each county to find the poverty population in each county
filtered_df["poverty_population"] = (filtered_df["totalpop"] * (filtered_df["pr_p"] / 100)).astype("int64")
filtered_df[["state_name","totalpop", "pr_p","poverty_population"]].head()
```

[490] 0.0s Python

	state_name	totalpop	pr_p	poverty_population
0	Alabama	55200	15.4	8500
1	Alabama	55380	15.2	8417
2	Alabama	55639	15.2	8457
3	Alabama	58239	13.6	7920
4	Alabama	58761	11.4	6698

- Dropping the general unemployment rate attribute (`unr_16`) from the table since the value can be derived from the gender-based unemployment rate attributes.

```
filtered_df = filtered_df.drop("unr_16", axis=1)
filtered_df.head(1)
```

[491] 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	funr_16	munr_16	mhi_2022	totalpop	pr_p	1_breadwinner_hh	2_breadwinners_hh	avg_ce
0	Alabama	AL	Autauga County	1001	2018	3.4	4.9	69367	55200	15.4	6127	5523	

- The attributes in the table below will be renamed for clarity.

Current Name	New Name
funr_16	female_unemployed_rate
munr_16	male_unemployed_rate
mhi_2022	2022_med_household_inc
pr_p	poverty_rate

```
#renaming some of the attributes for clarity
filtered_df = filtered_df.rename({
    "totalpop": "population",
    "funr_16": "female_unemployed_rate",
    "munr_16": "male_unemployed_rate",
    "mhi_2022": "2022_med_household_inc",
    "pr_p": "poverty_rate",
}, axis=1)

filtered_df.head(3)
```

[492] 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	female_unemployed_rate	male_unemployed_rate	2022_med_household_inc	population	pover
0	Alabama	AL	Autauga County	1001	2018	3.4	4.9	69367	55200	
1	Alabama	AL	Autauga County	1001	2019	3.2	4.1	68128	55380	
2	Alabama	AL	Autauga County	1001	2020	2.4	3.3	66679	55639	

2.3) Checking the Cleaned Dataset

```python
#checking the new shape of the final dataset
filtered_df.shape
```

[493] ✓ 0.0s Python

```
(16103, 15)
```

```python
filtered_df.info()
```

[494] ✓ 0.0s Python

```
<class 'pandas.DataFrame'>
RangeIndex: 16103 entries, 0 to 16102
Data columns (total 15 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   state_name                16103 non-null  str
 1   state_abbreviation        16103 non-null  str
 2   county_name               16103 non-null  str
 3   county_fips_code          16103 non-null  int64
 4   studyyear                 16103 non-null  int64
 5   female_unemployed_rate    16103 non-null  float64
 6   male_unemployed_rate      16103 non-null  float64
 7   2022_med_household_inc    16103 non-null  int64
 8   population                16103 non-null  int64
 9   poverty_rate              16103 non-null  float64
 10  1_breadwinner_hh          16103 non-null  int64
 11  2_breadwinners_hh         16103 non-null  int64
 12  avg_centrebased_care_price  16103 non-null  float64
 13  avg_family_childcare_price  16103 non-null  float64
 14  poverty_population        16103 non-null  int64
dtypes: float64(5), int64(7), str(3)
memory usage: 1.8 MB
```

iv) Exporting the Cleaned Dataset Before Commencing Analysis

```python
#creating a reusable function that can be used to export dataset at anytime
def export_dataset(dataset, pathname):
    try:
        output_path = Path(pathname)

        # create folder(s) if they don't exist
        output_path.parent.mkdir(parents=True, exist_ok=True)

        # save file
        dataset.to_csv(output_path, index=False)
    except:
        print("Encountered error while exporting the dataset. Skipping this process.")
    else:
        print("Dataset exported successfully.")
```

[495] ✓ 0.0s Python

```python
export_dataset(filtered_df, "./dataset/clean/cleaned_relevant_attributes_NDCP2022_dataset.csv")
```

[496] ✓ 0.1s Python

```
Dataset exported successfully.
```

3.0) Exploratory Data Analysis of the Cleaned Dataset

3.1) General Summary Statistics of the Dataset

```
#describe the only the numerical data in the dataset
filtered_df.drop(["county_fips_code", "studyyear"], axis = 1).describe().T
```

[497] ✓ 0.0s Python

	count	mean	std	min	25%	50%	75%	max
female_unemployed_rate	16103.0	5.278271	3.449434	0.000000	3.300000	4.700000	6.400000	3.900000e+01
male_unemployed_rate	16103.0	5.777284	3.692781	0.000000	3.700000	5.200000	7.000000	4.650000e+01
2022_med_household_inc	16103.0	61618.497050	17608.384925	14013.000000	50924.000000	59926.000000	69707.500000	1.709350e+05
population	16103.0	102561.790536	328441.534201	48.000000	11137.500000	25931.000000	66979.500000	1.009805e+07
poverty_rate	16103.0	15.588089	7.985098	0.000000	10.500000	14.000000	18.500000	6.710000e+01
1_breadwinner_hh	16103.0	10670.253307	37195.054185	0.000000	1022.500000	2546.000000	6606.000000	1.153355e+06
2_breadwinners_hh	16103.0	9194.269515	27997.475443	0.000000	876.000000	2126.000000	5788.500000	7.647500e+05
avg_centrebased_care_price	16103.0	151.683741	47.871146	23.558182	119.028182	144.744586	169.957273	5.238682e+02
avg_family_childcare_price	16103.0	126.636543	36.222570	44.312727	101.275455	124.335106	142.661364	4.035791e+02
poverty_population	16103.0	13765.451655	46120.342047	0.000000	1630.500000	4073.000000	10377.500000	1.615688e+06

Insights

- For both genders the poverty rates seem to have some extreem values on the high-end as **the mean and medium are around 5% poverty rate**, 75% of the population has below 7% poverty rate, **but the maxium is above 35%**. Though the extreem values are outliers, the values are essential within the scope of this analysis project and will be left as is.
- The standard deviation of the 2022 median household income show a high difference in median-household-incomes in the United States.
- On the poverty rate, the 75% of the popluation is below 18.5% but there are some outliers with poverty rate above 65%. These outliers will also be left as is for analysis.
- Based on the poverty rate, it appears that the county with the minimum poverty population has 13,765 people below the poverty line, while the highest has 1.6 million people below the poverty line. The average poverty population in the country is 13,765.
- The standard deviation of the centre-based and family childcare prices when compared to the mean shows a wider spread of differences in childcare accross the United States of America.
- Generally, there are no outlier values with invalid amounts e.g. negative amounts.

```
#setting the matplotlib color styles
sns.set_palette("Set2")
```

[498] ✓ 0.0s Python

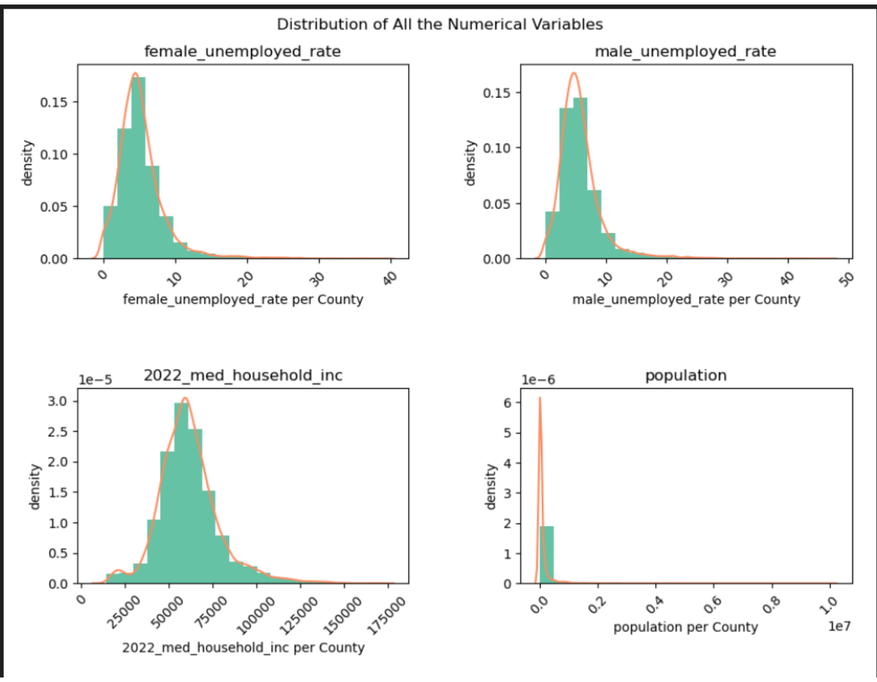
3.2) Checking the Distribution of each Variable and Showing Outliers

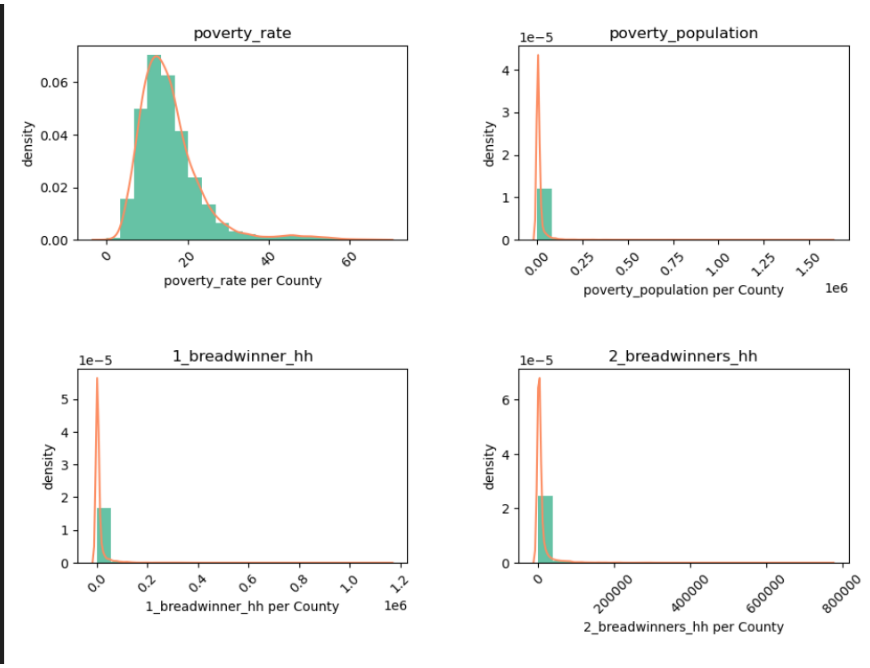
Outliers will be handled differently in this analysis because the goal is to analyse the real national records across the States. However, each of the numerical attributes will be visualised using a histogram to show the spread and how extreme the outliers are compared to the rest if there are visible outliers at all.

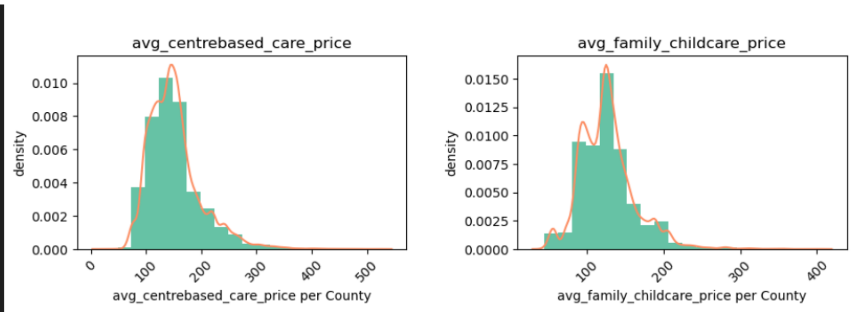
```
numeric_attr = ["female_unemployed_rate", "male_unemployed_rate", "2022_med_household_inc", "population", "poverty_rate", "poverty_population", "1_breadwinne
curr_col = 1
cols = 2
rows = int(len(numeric_attr)/cols)
rows += 1 if rows * cols < len(numeric_attr) else rows #if the number of rows is not enough, add 1 to it (it means the list of plots needed is odd not even)

plt.figure(figsize=(10, 35))
for attribute in numeric_attr:
    plt.subplot(rows, cols, curr_col)
    plt.title(attribute)
    plt.hist(filtered_df[attribute], bins=20, density=True)#ploting with density instead of count
    sns.kdeplot(filtered_df[attribute])
    plt.xlabel(f"{attribute} per County")
    plt.xticks(rotation = 45)
    plt.ylabel("density")
    curr_col += 1
plt.suptitle("Distribution of All the Numerical Variables")
plt.tight_layout(pad=3)
plt.show()
```

[499] ✓ 4.8s Python







From the histogram and kernel density plots above, most of the attributes have a normal distribution but are mostly positively skewed with a few having some high outliers e.g. `unemployment_rate`, `poverty_rate`, and `population` have majority of their samples being on the lesser side, while having some extreemly high outlier state counties.

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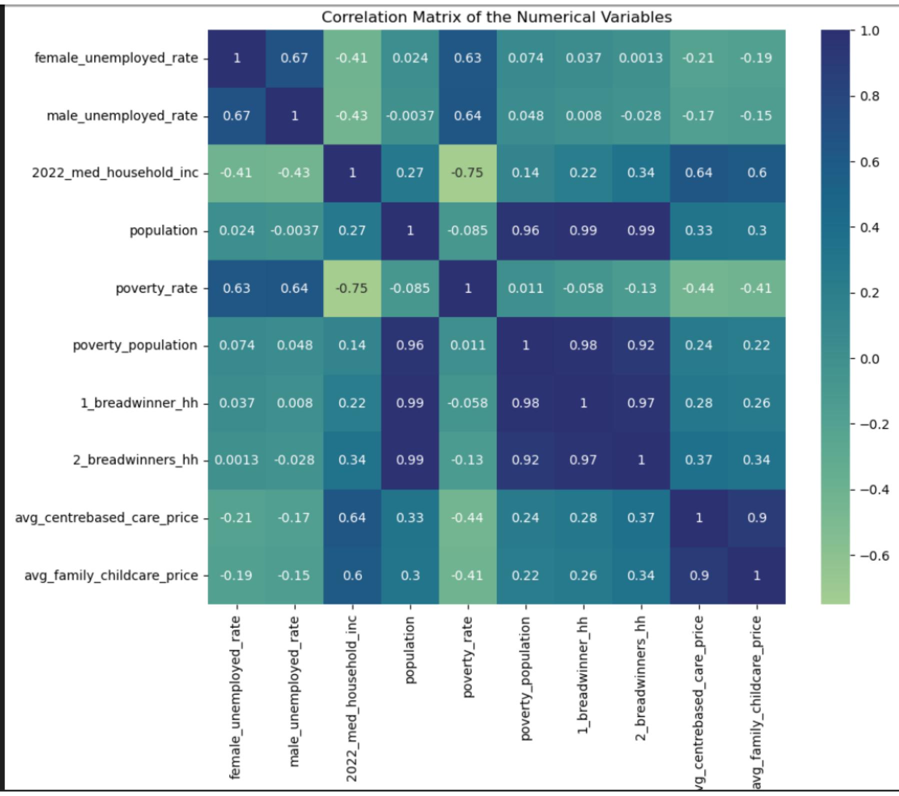
3.3) Analysing the Correlation Between all the Variables

Analysing the relationships between the numerical variables using correlation matrix

```
plt.figure(figsize=(10,8))
sns.heatmap(filtered_df[numeric_attr].corr(), cmap="crest", annot=True)
plt.title("Correlation Matrix of the Numerical Variables")
plt.show()
```

[500] ✓ 0.7s Python



The correlation matrix shows that:

- Average centre-based care prices correlate with the price of family shildcare prices accross most of the state counties with 90% correlation.
- The child-care prices have around 60% correlation with the median houshold income in each state county.
- The poverty rate in each state county is 63% correlated with the female unemployment rate and 64% correlated with the male unemployment rate.
- The poverty population has 24% correlation with centre-based care price and 22% correlation with family childcare price.
- The population size has no correlation with the unemployment rates.

3.4) Analysing the Unemployment Rate Variable as a Key Determinant

Checking the counties with the lowest unemployment rate

```
) == 0][["state_name", "county_name", "female_unemployed_rate", "male_unemployed_rate"]].groupby(by=["state_name", "county_name"], as_index=False).mean()
```

[501] ✓ 0.0s Python

	state_name	county_name	female_unemployed_rate	male_unemployed_rate
0	Colorado	Kiowa County	0.0	0.0
1	Hawaii	Kalawao County	0.0	0.0
2	Kansas	Greeley County	0.0	0.0
3	Kansas	Hamilton County	0.0	0.0
4	Montana	Fallon County	0.0	0.0
5	Montana	Garfield County	0.0	0.0
6	Montana	Liberty County	0.0	0.0
7	Montana	Treasure County	0.0	0.0
8	Montana	Wibaux County	0.0	0.0
9	Nebraska	Blaine County	0.0	0.0
10	Nebraska	Garfield County	0.0	0.0
11	Nebraska	Keya Paha County	0.0	0.0
12	Nebraska	Logan County	0.0	0.0

13	Nebraska	Loup County	0.0	0.0
14	Nebraska	McPherson County	0.0	0.0
15	Nebraska	Thomas County	0.0	0.0
16	Nebraska	Wheeler County	0.0	0.0
17	Nevada	Eureka County	0.0	0.0
18	New Mexico	Catron County	0.0	0.0
19	South Dakota	Haakon County	0.0	0.0
20	South Dakota	Hand County	0.0	0.0
21	South Dakota	Jerauld County	0.0	0.0
22	South Dakota	Stanley County	0.0	0.0
23	South Dakota	Sully County	0.0	0.0
24	Texas	Crockett County	0.0	0.0
25	Texas	Donley County	0.0	0.0
26	Texas	Edwards County	0.0	0.0
27	Texas	Glasscock County	0.0	0.0
28	Texas	Jeff Davis County	0.0	0.0
29	Texas	Kenedy County	0.0	0.0
30	Texas	Loving County	0.0	0.0
31	Texas	Roberts County	0.0	0.0
32	Texas	Sterling County	0.0	0.0
33	Texas	Terrell County	0.0	0.0
34	Utah	Rich County	0.0	0.0

Checking the county with the highest male unemployment rate

Generate + Code + Markdown

```python
#Checking the county with the highest male unemployment rate
filtered_df[filtered_df["male_unemployed_rate"] == filtered_df["male_unemployed_rate"].max()]
```

[502] ✓ 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	female_unemployed_rate	male_unemployed_rate	2022_med_household_inc	population	p
15855	Puerto Rico	PR	Guanica County	72055	2020	20.3	46.5	14125	15825	

Checking the county with the highest female unemployment rate

```python
#Checking the county with the highest female unemployment rate
filtered_df[filtered_df["female_unemployed_rate"] == filtered_df["female_unemployed_rate"].max()]
```

[503] ✓ 0.0s Python

	state_name	state_abbreviation	county_name	county_fips_code	studyyear	female_unemployed_rate	male_unemployed_rate	2022_med_household_inc	population	p
16070	Puerto Rico	PR	Utuado County	72141	2020	39.0	27.9	17274	28041	

Viewing the anual trend of unemployment rate in the united states

```python
#extracting a subset of the dataset to use to further analyse the anual unemployment rate
annual_unemp = filtered_df[["studyyear", "female_unemployed_rate", "male_unemployed_rate"]]
annual_unemp["mean_unemployment"] = ((annual_unemp["female_unemployed_rate"] + annual_unemp["male_unemployed_rate"])/2).round(2)
annual_unemp.head()
```

[504] ✓ 0.0s Python

	studyyear	female_unemployed_rate	male_unemployed_rate	mean_unemployment
0	2018	3.4	4.9	4.15
1	2019	3.2	4.1	3.65
2	2020	2.4	3.3	2.85
3	2021	2.3	3.3	2.80
4	2022	2.6	2.9	2.75

```python
#showing a summary of the average unemployment per year
annual_unemp.groupby(by = "studyyear").mean()
```

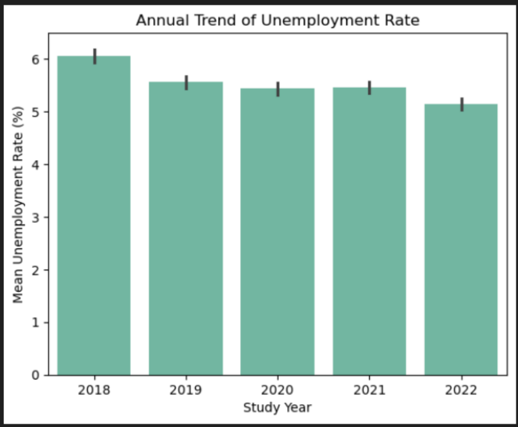
[505] ✓ 0.0s Python

	female_unemployed_rate	male_unemployed_rate	mean_unemployment
studyyear			
2018	5.788075	6.324130	6.056102
2019	5.288385	5.825093	5.556739
2020	5.212760	5.659578	5.436169
2021	5.199224	5.704626	5.451925
2022	4.903074	5.373176	5.138125

Visualising the summary of unemployment rate per year showing the trend

```python
sns.barplot(data=annual_unemp, x="studyyear", y="mean_unemployment")
plt.title("Annual Trend of Unemployment Rate")
plt.xlabel("Study Year")
plt.ylabel("Mean Unemployment Rate (%)")
plt.show()
```

[506] ✓ 0.6s Python

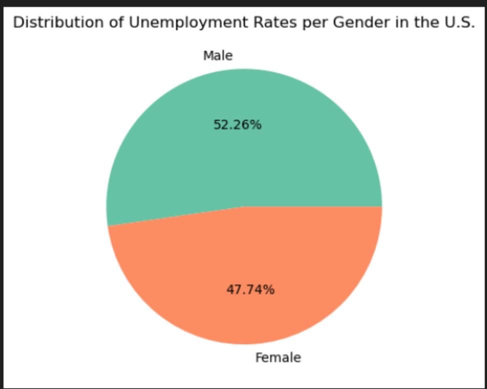


The bar chart above shows a gradual decrease in the annual unemployment rate in the United States from 2018 to 2022. The lowest being 2022 with unemployment rate of 5.13%

For each gender, visualising the average unemployment rate for both genders

```python
unemployent_rate_male = annual_unemp["male_unemployed_rate"].mean() #this is of 100%
unemployent_rate_female = annual_unemp["female_unemployed_rate"].mean() #this is of 100%

plt.pie([unemployent_rate_male, unemployent_rate_female], labels=["Male", "Female"], autopct="%.2f%%")
plt.title("Distribution of Unemployment Rates per Gender in the U.S.")
plt.show()
```

[507] ✓ 0.1s Python



Insights summary:

- About 35 counties are recorded to have a 0% unemployment rate.
- The Guanica County has the highest male unemployment rate of 46.5% and the Utuado County has the highest female unemployment rate of 39% in 2020. Both counties are in Puerto Rico.
- Nationally, there was a gradual decrease in the annual unemployment rate in the United States from 2018 to 2022. The lowest being 2022 with unemployment rate of 5.13%.
- The piechart shows that males have higher unemployment rate than females in the United States. 52.26% and 47.74% respectively.

3.5) Analysing the relationships between the other variables with high correlation scores

Visualising a some of the variables that show correlation for each State County

1. centre-based care prices against family child-care prices

```
def visualiseReationship(x, y, xlabel, ylabel, data_label = None, max_annot_text = "", marker = ".", transparent = False, size = 20):
    #scale the variables around their mean
    x_norm = (x - x.min()) / (x.max() - x.min()) #normalise the values of x
    y_norm = (y - y.min()) / (y.max() - y.min()) #normalise the values of y
    opacity = 1 if transparent == False else 0.6

    plt.scatter(x_norm, y_norm, marker=marker, label=data_label, alpha=opacity, s=size)

    if max_annot_text:
        plt.annotate(max_annot_text, xy = (x_norm.max()+0.03, y_norm.max()-0.01), bbox=dict(boxstyle="round", fc=(1.0, 0.7, 0.7), ec="none"))
    plt.xlabel(f"{xlabel} (Normalised)")
    plt.ylabel(f"{ylabel} (Normalised)")
```

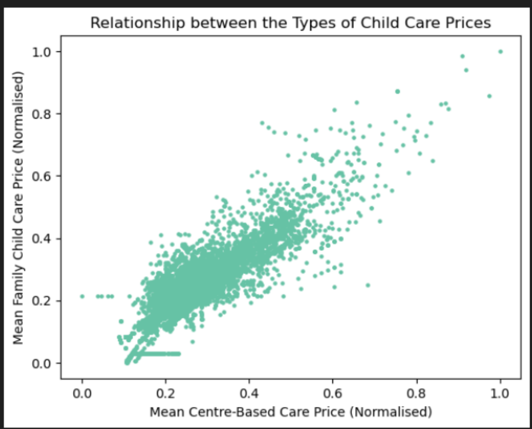
[508] ✓ 0.0s Python

```
def visualiseReationshipKDE(x, y, xlabel, ylabel, max_annot_text = "", color = None):
    #scale the variables around their mean
    x_norm = (x - x.min()) / (x.max() - x.min()) #normalise the values of x
    y_norm = (y - y.min()) / (y.max() - y.min()) #normalise the values of y

    sns.kdeplot(x = x_norm, y = y_norm, fill=True, color=color)

    if max_annot_text:
        plt.annotate(max_annot_text, xy = (x_norm.max()+0.03, y_norm.max()-0.01), bbox=dict(boxstyle="round", fc=(1.0, 0.7, 0.7), ec="none"))
    plt.xlabel(f"{xlabel} (Normalised)")
    plt.ylabel(f"{ylabel} (Normalised)")
```

[509] ✓ 0.0s Python

```
visualiseReationship(filtered_df["avg_centrebased_care_price"], filtered_df["avg_family_childcare_price"], xlabel="Mean Centre-Based Care Price", ylabel="Mea
plt.title("Relationship between the Types of Child Care Prices")
plt.show()
```

[510] ✓ 0.3s Python

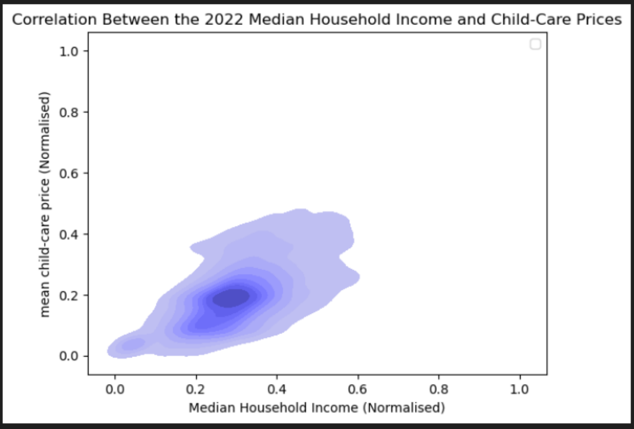


Mean Centre-Based Care price and the Family Child-Care price are proportional to each-other, hence, there is a linear relationship between the two. An increase in one mostly lead to an increase in the other.

2. Relationship Between the 2022 Median Household Income and Child-Care Price

```
avg_chilcare_price_series = (filtered_df[["avg_centrebased_care_price","avg_family_childcare_price"]]).mean(axis=1)
visualiseReationshipKDE(x = filtered_df["2022_med_household_inc"], y = avg_chilcare_price_series, xlabel="Median Household Income", ylabel="mean child-care
plt.title("Correlation Between the 2022 Median Household Income and Child-Care Prices")
plt.legend()
plt.show()
```

[511] ✓ 27.1s Python

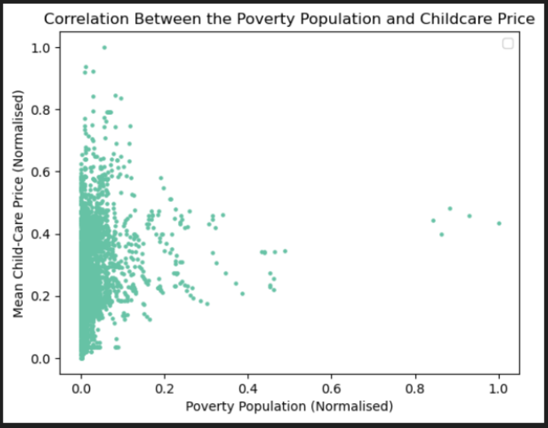


The two dimensional Kernel Density Estimation (KDE) plot visualisation shows a linear relationship between Median Household Income and Mean Child-Care Price in each County.

3. Visualising the relationship between **Poverty Population** and **Average Child-Care Prices** in each county

```
visualiseReationship(x = filtered_df["poverty_population"], y = avg_chilcare_price_series, xlabel="Poverty Population", ylabel="Mean Child-Care Price")
plt.title("Correlation Between the Poverty Population and Childcare Price")
plt.legend()
plt.show()
```

[512] ✓ 0.2s Python



As seen above, the majority of the State Counties have poverty population on the lower end except some outliers and the relationship between the poverty population and the child-care price does not appear to have a linear shape as there is high variation of prices across the State Counties.

4.0) In-Depth Multivariate Data Analysis

Primary Objectives of this Section

1. Analyse the trend of average U.S unemployment rate by gender against the annual poverty rate across the years from 2018 to 2022.
2. Analyse the relationship between unemployment rate and poverty rate per gender per state.
3. Create choropleth visualisation of median household income and poverty rate across states.
4. Analyse the effects of proportion of households with only one working parent on the poverty rate in the states.
5. Analyse the relationship between average weekly child-care costs and poverty rates across states.

4.1) Analysing the Trend of Average U.S Unemployment Rate by Gender Against the Annual Poverty Rate Across the Years from 2018 to 2022

Firstly, summarising the unemployment rate and poverty rate per gender per year

```python
mean_annual_unemp_rate = filtered_df[["studyyear", "male_unemployed_rate", "female_unemployed_rate"]].groupby("studyyear", as_index=False).mean()
mean_annual_unemp_rate = mean_annual_unemp_rate.rename({ #renaming this coulmn so the data label comes out clearer on the chart
    "studyyear": "Study Year",
    "male_unemployed_rate": "male unemployment rate",
    "female_unemployed_rate": "female unemployment rate"
}, axis=1)

#convert the table from wide table to long table such that there will be a new column containing the Gender and the second column being their corresponding u
mean_annual_unemp_rate = mean_annual_unemp_rate.melt(id_vars="Study Year", var_name="Gender", value_name="Unemployment Rate")
mean_annual_unemp_rate.head()
```

[513] 0.0s Python

	Study Year	Gender	Unemployment Rate
0	2018	male unemployment rate	6.324130
1	2019	male unemployment rate	5.825093
2	2020	male unemployment rate	5.659578
3	2021	male unemployment rate	5.704626
4	2022	male unemployment rate	5.373176

Seperating the male and female annual average unemployment rates into seperate dataframes for easier visualisation with multiple bar chart

```python
#seperating the male and female unemployment rates into seperate dataframes
male_annual_uemp_rate = mean_annual_unemp_rate[mean_annual_unemp_rate["Gender"] == "male unemployment rate"]
female_annual_uemp_rate = mean_annual_unemp_rate[mean_annual_unemp_rate["Gender"] == "female unemployment rate"]

print("Female Unemployment Rate (Sorted by Study Year)")
female_annual_uemp_rate #previewing the female table
```

[514] 0.0s Python

```
Female Unemployment Rate (Sorted by Study Year)
```

	Study Year	Gender	Unemployment Rate
5	2018	female unemployment rate	5.788075
6	2019	female unemployment rate	5.288385
7	2020	female unemployment rate	5.212760
8	2021	female unemployment rate	5.199224
9	2022	female unemployment rate	4.903074

Aggregating the poverty rate per year

```python
#first aggregating the total population and total poverty population per year
annual_aggregate_df = filtered_df[["studyyear","population", "poverty_population"]].groupby("studyyear", as_index=False).sum()

#calculating the annual poverty rate
annual_aggregate_df["poverty_rate(%)"] = ((annual_aggregate_df["poverty_population"] / annual_aggregate_df["population"]) * 100).round(2)
annual_aggregate_df.head()
```

[515] 0.0s Python

	studyyear	population	poverty_population	poverty_rate(%)
0	2018	326289971	46967586	14.39
1	2019	328016242	45136179	13.76
2	2020	329824950	43443082	13.17
3	2021	333036755	43156876	12.96
4	2022	334384595	42961345	12.85

Analysing the year-on-year rate of change of the unemployment rate and poverty rate

```
print("The year-on-year difference in male unemployment rate in the US from 2018 to 2022 is: ", end="")
print(f"{male_annual_uemp_rate.sort_values(by = "Study Year")["Unemployment Rate"].diff().mean().__round__(2)}%")
```

[516] ✓ 0.0s Python

```
The year-on-year difference in male unemployment rate in the US from 2018 to 2022 is: -0.24%
```

```
print("The year-on-year difference in female unemployment rate in the US from 2018 to 2022 is: ", end="")
print(f"{female_annual_uemp_rate.sort_values(by = "Study Year")["Unemployment Rate"].diff().mean().__round__(2)}%")
```

[517] ✓ 0.0s Python

```
The year-on-year difference in female unemployment rate in the US from 2018 to 2022 is: -0.22%
```

```
print("The year-on-year difference in poverty rate in the US from 2018 to 2022 is: ", end="")
print(f"{annual_aggregate_df.sort_values(by = "studyyear")["poverty_rate(%)"].diff().mean().__round__(2)}%")
```

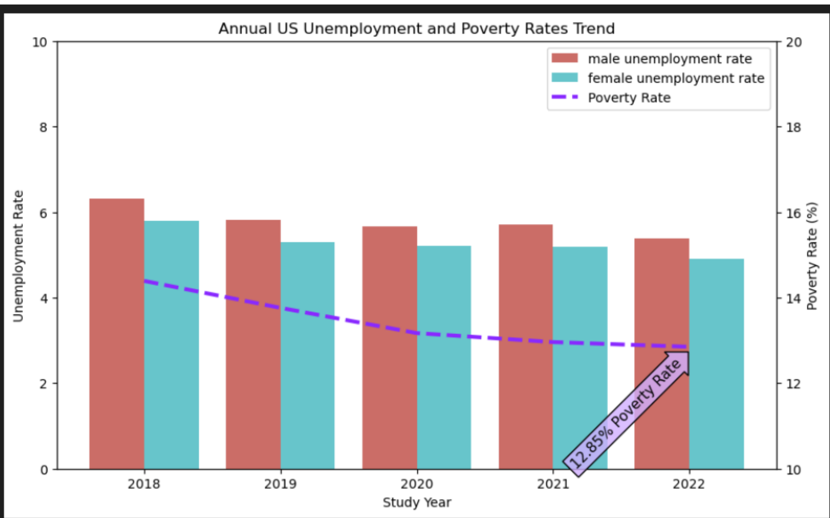
[518] ✓ 0.0s Python

```
The year-on-year difference in poverty rate in the US from 2018 to 2022 is: -0.39%
```

Visualising the annual unemployment rate by gender and poverty rate to find patterns

```
#Draw a grouped bar chart Visualisations to visualise the annual unemployment rate per gender
fig, ax1 = plt.subplots(figsize=(10,6))
sns.barplot(data = mean_annual_unemp_rate, x = "Study Year", y = "Unemployment Rate", hue = "Gender", ax=ax1, palette= "hls") #use seaborn to plot the line
ax1.set_ylim(bottom = 0, top=10)
handles1, label1 = ax1.get_legend_handles_labels()

#Plot the average poverty rate over the years
ax2 = ax1.twinx()
ax2.plot(annual_aggregate_df["poverty_rate(%)"], c = "#8929ff", linestyle = "dashed", linewidth=3.0, label = "Poverty Rate")
ax2.set_ylabel("Poverty Rate (%)")
ax2.set_ylim(bottom = 10, top = 20)
handles2, label2 = ax2.get_legend_handles_labels()

plt.text(3.54, annual_aggregate_df["poverty_rate(%)"].max() - 3.1, f"{round(annual_aggregate_df["poverty_rate(%)"].min(), 2)}% Poverty Rate",
         ha="center", va="center", rotation=45, size=11,
         bbox=dict(boxstyle="rarrow,pad=0.3",
                   fc=(0.82, 0.69, 1, 0.8), lw=1))

ax1.legend(handles= handles1 + handles2, labels= label1 + label2)
plt.title("Annual US Unemployment and Poverty Rates Trend")
plt.show()
```

[519] ✓ 0.3s Python



Insights:
There was a steady decline of average unemployment rate in the United States across both genders from 2018 to 2022 with a year-on-year decrease of -0.23%, the poverty rate also had a yearly decline rate with almost the same magnitude with a year-on-year decrease of -0.39%. The lowest being 2022 that has poverty rate of 12.85%, male unemployment rate of 5.3% and female unemployment rate is 4.9%. The visualisation above showed a correlation in the trends of poverty rate and unemployment rate over the years.

4.2) Analysing the Relationship Between Unemployment Rate and Poverty Rate per Gender per State

Aggregating the 2022 State-Level Data for Further Analaysis

Since the next couple of objectives are state-level analysis, the state-level dataset will be generated by aggregating the data from the dataset grouping by States.

```
d using the mean and group the rest by the state name and other variables using the mean
population", "poverty_rate", "county_name","county_fips_code","1_breadwinner_hh","2_breadwinners_hh"], axis=1).groupby(["state_name","state_abbreviation"], a
d the breadwinner-type variables by summing each of them
ion", "poverty_population", "1_breadwinner_hh","2_breadwinners_hh"]].groupby("state_name", as_index=False).sum()
overty rate
ns["poverty_population"] / states_populations["population"]) * 100).round(2)

tes data
="inner")


22_med_household_inc"].round(2) #round up the medium household income to 2 decimal places
ns for clearer labeling in the map
($)",
```

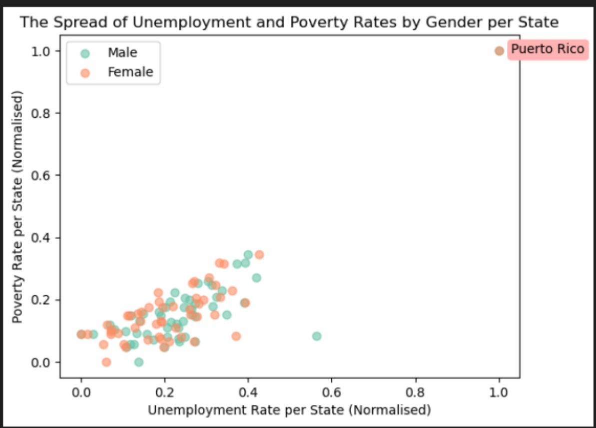
[520] ✓ 0.0s Python

	State Name	Abbreviation	studyyear	female_unemployed_rate	male_unemployed_rate	Median Household Income ($)	avg_centrebased_care_price	avg_family_childcare_price	population	pov
0	Alabama	AL	2020.00000	6.685075	6.739701	51052.31	119.086098	109.243541	24659883	
1	Alaska	AK	2020.02027	7.324324	10.035135	78169.14	242.180721	204.839633	3683346	
2	Arizona	AZ	2020.00000	7.620000	7.761333	58130.81	165.818570	139.513855	35422533	
3	Arkansas	AR	2020.00000	5.941867	6.296000	49349.06	103.929326	91.314389	15026892	
4	California	CA	2020.00000	6.634828	7.196552	80397.23	237.930513	205.329726	196589737	

Performing the Analysis of Unemployment Rate per Gender per State

```
male_data = states_data[["State Name", "Poverty Rate (%)", "male_unemployed_rate"]]
female_data = states_data[["State Name", "Poverty Rate (%)", "female_unemployed_rate"]]
m_max_annotate_text = male_data[male_data["Poverty Rate (%)"] == male_data["Poverty Rate (%)"].max()]["State Name"].iloc[0] #to show the name of the state wi
f_max_annotate_text = female_data[female_data["Poverty Rate (%)"] == female_data["Poverty Rate (%)"].max()]["State Name"].iloc[0] #to show the name of the st

visualiseReationship(male_data["male_unemployed_rate"], male_data["Poverty Rate (%)"], data_label="Male", ylabel="Poverty Rate per State", xlabel="Unemployme
visualiseReationship(female_data["female_unemployed_rate"], female_data["Poverty Rate (%)"], data_label="Female", ylabel="Poverty Rate per State", xlabel="Un
plt.title("The Spread of Unemployment and Poverty Rates by Gender per State")
plt.legend()
plt.show()
```

[521] ✓ 0.2s Python



Showing the details of the top 5 States with the highest Unemployment Rate

```
states_data["avg_unemployment_rate"] = ((male_data["male_unemployed_rate"] + female_data["female_unemployed_rate"])/2).round(2) #get the average unemploymen
states_data[["State Name", "avg_unemployment_rate"]].sort_values(by = "avg_unemployment_rate", ascending=False).head()
```

[522] ✓ 0.0s Python

	State Name	avg_unemployment_rate
39	Puerto Rico	15.81
1	Alaska	8.68
24	Mississippi	7.98
2	Arizona	7.69
49	West Virginia	7.30

Showing the details of the top 5 States with the highest Poverty Rate

```
states_data[["State Name", "Poverty Rate (%)"]].sort_values(by = "Poverty Rate (%)", ascending=False).head()
```

[523] ✓ 0.0s Python

	State Name	Poverty Rate (%)
39	Puerto Rico	43.38
24	Mississippi	19.93
18	Louisiana	19.00
31	New Mexico	18.87
49	West Virginia	17.23

Insights:
Down to the States level, the scatter plot further shows the spread and the general linear relationship between unemployment rate and poverty rates in all the States. The analysis also spotlights the State of Puerto Rico which has the highest unemployment and poverty rate in the United State, 15.81% and 43.38% respectively.

4.3) Creating Interactive Choropleth Visualisation of Median Household Income and Poverty Rate Across States

Visualising using an interactive Pyplot choropleth to show the `States`, the `Median Household Income` and `Poverty Rates`. The Pyplot figure will be drawn using shape-points GeoJSON data of US States publically available and downloaded from https://raw.githubusercontent.com/PublicaMundi/MappingAPI/master/data/geojson/us-states.json

```
import json

#importing the full USA states GeoJSON shape data, this will enable Plotly render the shapes of all the states including Puerto-Rico
with open("./dataset/us-states.geojson", "r") as gj:
    geojson_data = json.load(gj)
```

[524] ✓ 0.0s Python

Adding GeoLocation Centroids (Lon/Lat) to the States Dataset
Using the US States' longitude and latitude centroids dataset publically available at https://developers.google.com/public-data/docs/canonical/states_csv

```
states_centroids = pd.read_csv("./dataset/us-states-lon-lat-centroids.csv") #importing the downloaded centroids data
states_centroids.head()
```

[525] ✓ 0.0s Python

	state_abr	latitude	longitude	state_name
0	AK	63.588753	-154.493062	Alaska
1	AL	32.318231	-86.902298	Alabama
2	AR	35.201050	-91.831833	Arkansas
3	AZ	34.048928	-111.093731	Arizona
4	CA	36.778261	-119.417932	California

```
#merging the centroid geo-location data into the states dataset
states_data_and_loc = states_data.merge(states_centroids, left_on="Abbreviation", right_on="state_abr")
states_data_and_loc = states_data_and_loc.drop(["state_abr", "state_name"], axis = 1) #drop the duplicate columns
states_data_and_loc.head()
```

[526] ✓ 0.0s Python

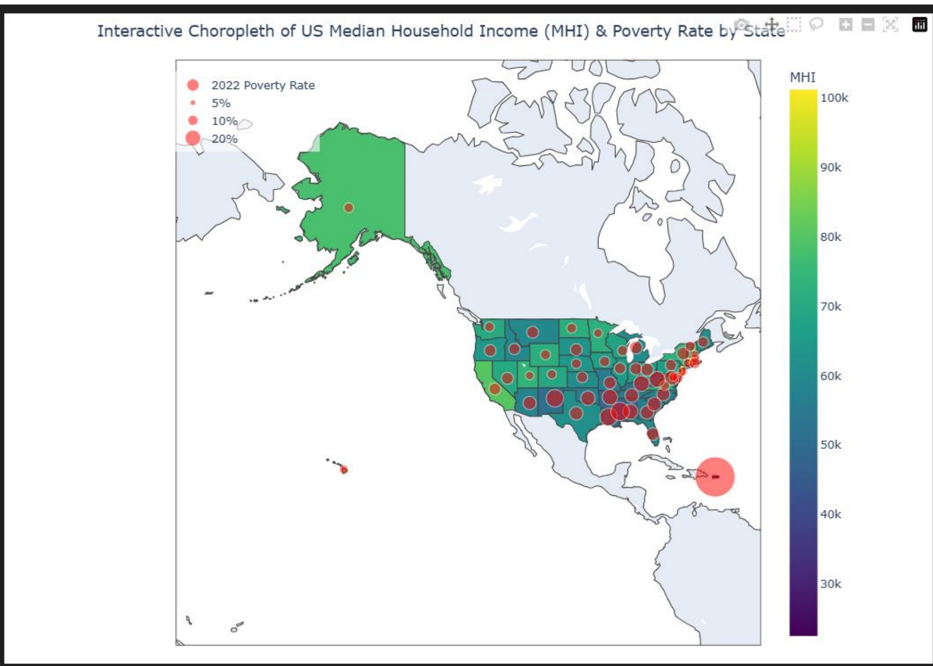
	State Name	Abbreviation	studyyear	female_unemployed_rate	male_unemployed_rate	Median Household Income ($)	avg_centrebased_care_price	avg_family_childcare_price	population	pov
0	Alabama	AL	2020.00000	6.685075	6.739701	51052.31	119.086098	109.243541	24659883	
1	Alaska	AK	2020.02027	7.324324	10.035135	78169.14	242.180721	204.839633	3683346	
2	Arizona	AZ	2020.00000	7.620000	7.761333	58130.81	165.818570	139.513855	35422533	
3	Arkansas	AR	2020.00000	5.941867	6.296000	49349.06	103.929326	91.314389	15026892	
4	California	CA	2020.00000	6.634828	7.196552	80397.23	237.930513	205.329726	196589737	

Visualising the Median Household Income and Poverty Rate per State using Choropleth

```python
# plot choropleth with Plotly
fig = go.Figure()

fig.add_trace(go.Choropleth(
    geojson=geojson_data,
    featureidkey="properties.name",
    locations=states_data_and_loc["State Name"],
    z=states_data_and_loc["Median Household Income ($)"],
    colorscale="Viridis",
    name="Household Income",
    # hovertext=states_data_and_loc["State Name"],
    colorbar=dict(
        title="MHI",
        x=0.85
    )
))

# plotting the poverty rate per state using a bubble marker on the choropleth
fig.add_trace(go.Scattergeo(
    lon=states_data_and_loc["longitude"],
    lat=states_data_and_loc["latitude"],
    text=states_data_and_loc["State Name"] + "<br>Poverty Rate: " + states_data_and_loc["Poverty Rate (%)"].astype(str) +"%<br>Median Houshold Income: $" + s
    mode="markers",
    marker=dict(
        size=states_data_and_loc["Poverty Rate (%)"],
        color="red",
        opacity=0.5,
        line=dict(width=1, color="white")
    ),
    name="2022 Poverty Rate"
))
```

```python
#drawing the bubble chart's legend
legend_bubble_sizes = [5, 10, 20]
for s in legend_bubble_sizes:
    fig.add_trace(go.Scattergeo(
        lon=[None],
        lat=[None],
        mode="markers",
        marker=dict(
            size= s,
            color="red",
            opacity=0.5
        ),
        name=f"{s}%"
))
#------------------

fig.update_geos(
    fitbounds="locations", #fit the chart to view all locations
    projection_type="mercator", #the map was looking stretched so this makes it appear similar to how maps are drawn generally
)

fig.update_layout(
    width=1000,
    height=700,
    margin=dict(l=20, r=20, t=50, b=20),
    legend=dict( #move the bubbles legend to the left
        x=0.17,
        y=0.98,
        bgcolor="rgba(255,255,255,0.6)"
    ),
    title=dict(
        text = "Interactive Choropleth of US Median Household Income (MHI) & Poverty Rate by State",
        x = 0.1
    )
)

fig.show()
```

[527] ✓ 0.1s Python



INSIGHTS:

- The combined interactive choropleth and bubble map visualisation above, shows the spread of the median household income across the states and compares them with their poverty rates.
- From the visualisation, it is observed that many of the southern States in the US have averagely lower median household income compared to the other regions. It also showed that the southern states averagely have higher poverty rates than other regions.
- The States with the highest median household income are in the eastern region towards the shores of the Country.
- The visualisation emphasises the extreme outlier in the south; the **State of Puerto Rico** with the highest poverty rate of 43.38% and the lowest median household income of $22,524.21 in the country.

4.4) Analysing the Proportion of one-breadwinner households (with kids below 18 years) and the impact on the States' poverty rates

Calculating the ratio of the two classes of households (between one bread winner and two bread winners households)

```
household_ratio_data = pd.DataFrame(states_data["State Name"])
household_ratio_data["ratio_1_breadwinner_hh(%)"] = ((states_data["1_breadwinner_hh"] / states_data[["1_breadwinner_hh", "2_breadwinners_hh"]].sum(axis=1))
household_ratio_data.head()#checking the resulting dataset
```

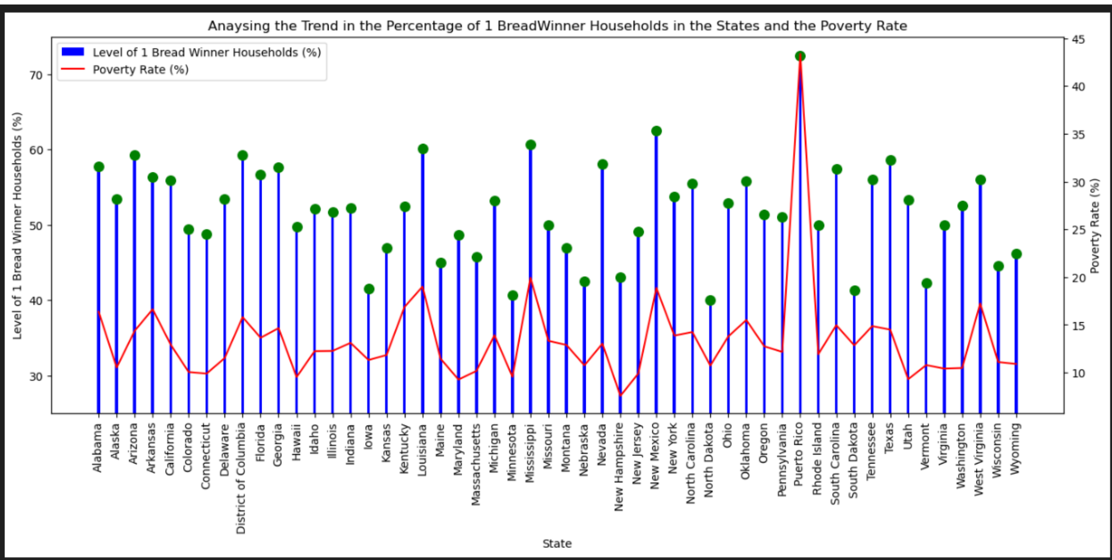
[528] ✓ 0.0s Python

	State Name	ratio_1_breadwinner_hh(%)
0	Alabama	57.78
1	Alaska	53.46
2	Arizona	59.27
3	Arkansas	56.34
4	California	55.88

A) Households with Only Father Working

```
fig, ax1 = plt.subplots(figsize=(16,6))
ax1.bar(x = states_data["State Name"], height= household_ratio_data["ratio_1_breadwinner_hh(%)"], width=0.16, color="blue", label="Level of 1 Bread Winner Ho
ax1.scatter(x = states_data["State Name"], y = household_ratio_data["ratio_1_breadwinner_hh(%)"], s=70, c="green")
ax1.set_ylim(bottom=25, top=75)
ax1.set_xlabel("State")
ax1.set_ylabel("Level of 1 Bread Winner Households (%)")
handles1, label1 = ax1.get_legend_handles_labels()

plt.xticks(rotation = 90)

ax2 = ax1.twinx()
ax2.plot(states_data["Poverty Rate (%)"], color="red", label="Poverty Rate (%)")
ax2.set_ylabel("Poverty Rate (%)")
handles2, label2 = ax2.get_legend_handles_labels()

ax1.legend(handles= handles1 + handles2, labels= label1 + label2, loc="best")
plt.title("Anaysing the Trend in the Percentage of 1 BreadWinner Households in the States and the Poverty Rate")
plt.show()
```

[529] ✓ 0.7s Python



INSIGHTS:

The visualisation above shows a trend of poverty rates being proportional to the rate of one-breadwinner households in the States. Especially in Puerto-Rico where over 70% of households raising kids below the age of 18 has only one working parent.

4.5) Analysing effects of the States' average weekly child-care cost on the poverty rate

```
states_childcare_cost_df = states_data[["State Name", "avg_centrebased_care_price", "avg_family_childcare_price", "Poverty Rate (%)"]]
states_childcare_cost_df["avg_child_care_price"] = states_childcare_cost_df[["avg_centrebased_care_price", "avg_family_childcare_price"]].mean(axis = 1).rou
states_childcare_cost_df.head()
```

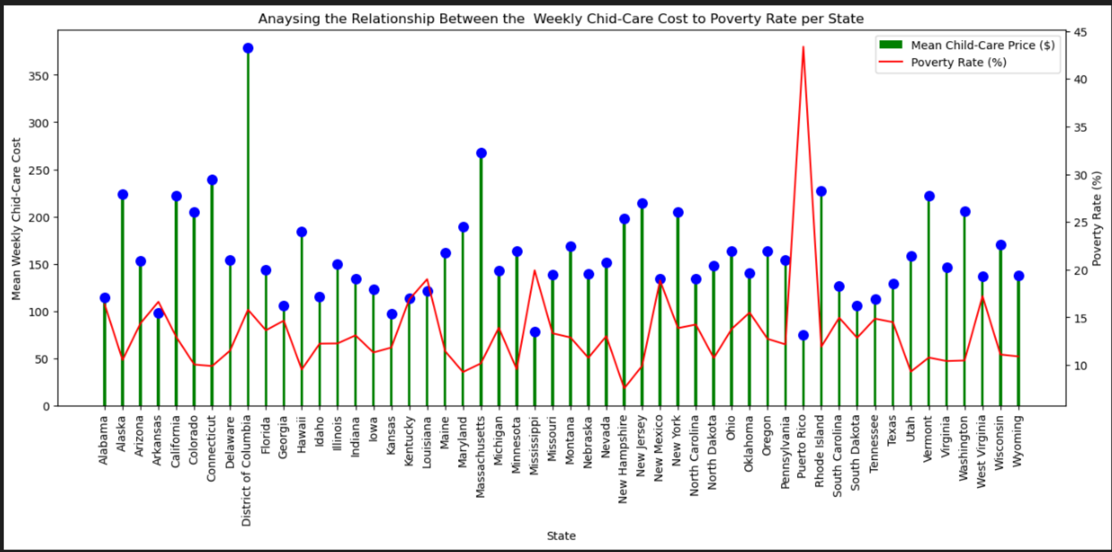
[530] ✓ 0.0s Python

	State Name	avg_centrebased_care_price	avg_family_childcare_price	Poverty Rate (%)	avg_child_care_price
0	Alabama	119.086098	109.243541	16.36	114.16
1	Alaska	242.180721	204.839633	10.52	223.51
2	Arizona	165.818570	139.513855	14.37	152.67
3	Arkansas	103.929326	91.314389	16.63	97.62
4	California	237.930513	205.329726	12.93	221.63

```
fig, ax1 = plt.subplots(figsize=(16,6))
ax1.bar(x = states_childcare_cost_df["State Name"], height= states_childcare_cost_df["avg_child_care_price"], width=0.16, color="green", label="Mean Child-Ca
ax1.scatter(x = states_childcare_cost_df["State Name"], y = states_childcare_cost_df["avg_child_care_price"], s=70, c="blue")
plt.xticks(rotation = 90)
# ax1.set_ylim(bottom=35, top=75)
ax1.set_xlabel("State")
ax1.set_ylabel("Mean Weekly Chid-Care Cost")
handles1, label1 = ax1.get_legend_handles_labels()

ax2 = ax1.twinx()
ax2.plot(states_childcare_cost_df["Poverty Rate (%)"], color="red", label="Poverty Rate (%)")
ax2.set_ylabel("Poverty Rate (%)")
handles2, label2 = ax2.get_legend_handles_labels()
```

```python
ax1.legend(handles= handles1 + handles2, labels= label1 + label2, loc="best")
plt.title("Anaysing the Relationship Between the  Weekly Chid-Care Cost to Poverty Rate per State")
plt.show()
```

[531] ✓ 0.8s Python



Analysing the States with the highest and lowest weekly child-care cost

```python
#showing the States with the highest child-care cost
states_childcare_cost_df.sort_values(by="avg_child_care_price", ascending=False).head(3)
```

[532] ✓ 0.0s Python

	State Name	avg_centrebased_care_price	avg_family_childcare_price	Poverty Rate (%)	avg_child_care_price
8	District of Columbia	400.777273	357.290909	15.80	379.03
21	Massachusetts	304.758078	229.799221	10.17	267.28
6	Connecticut	271.272727	207.852273	9.89	239.56

```python
#showing the States with the lowest child-care cost
states_childcare_cost_df.sort_values(by="avg_child_care_price", ascending=True).head(3)
```

[533] ✓ 0.0s Python

	State Name	avg_centrebased_care_price	avg_family_childcare_price	Poverty Rate (%)	avg_child_care_price
39	Puerto Rico	75.898189	74.169573	43.38	75.03
24	Mississippi	100.971162	55.417129	19.93	78.19
16	Kansas	104.978978	89.282424	11.83	97.13

INSIGHTS:

The chart above shows a very slight negative correlation trend between the average cost of child-care and the poverty rates in the States. The State with the highest weekly child-care cost of $379.03 was the **District of Columbia** but the poverty rate is 15.8% which is the national average. While the state with the lowest child-care cost is Puerto-Rico with an average child-care cost of $75.03 but has the highest poverty rate in the country.

5.0) General Conclusions Based on the Outcomes

The result of this analysis shows the true view of the wealth disparity in a first-world country like the United States of America. The results suggests that nation-wide averages of key socio-economic variables may not be fully representative of the realities within the Counties and the States in the US.

Considering the insights gained from this analysis, the following conclusions and recommendations are presented:

- There is a positive correlation between unemployment rate and poverty rate in the States, more Government initiatives for job creation may further help improve the socio-economic state in many regions in the US and reduce the poverty rate.
- The large income and poverty-rate gaps between the State of Puerto-Rico and other States may suggest an existence of low prioritisation in resource allocation and economic development planning for that state.
- Reforms in Government resource allocation may lead to better wealth distribution among the States, especially for States in the Southern region and most especially for the State of Puerto-Rico.
- Public awareness campaigns on dual-earner households could help reduce poverty levels in States that typically have households relying on one breadwinner.